

Simple Interest and Compound Interest

Simple Interest

If Principal = Rs. 'P', Time = 'T' years, Rate = 'R%' per annum,

$$\text{Simple Interest (SI)} = \frac{P \times R \times T}{100}$$

Amount = Principal + Simple Interest

$$A = P + \frac{P \times R \times T}{100} = P \left[1 + \frac{RT}{100} \right]$$

Example: Find the simple interest on Rs. 200 for 5 years at 6% per annum?

Sol. Here, P = Rs. 200, T = 5 years, R = 6%

$$\therefore SI = \frac{P \times R \times T}{100} = \frac{200 \times 5 \times 6}{100} = \text{Rs. } 60$$

(i) If rate of interest is half-yearly, Rate = $\left(\frac{R}{2}\right)\%$ and Time = 2T

(ii) If rate of interest is quarterly, Rate = $\left(\frac{R}{4}\right)\%$ and Time = 4T

(iii) If rate of interest is monthly, Rate = $\left(\frac{R}{12}\right)\%$ and Time = 12T

Installments

When the borrower paid total money in some equal parts, then we can say that he is paying in installments.

For simple interest,

$$A = \left[x + \left(x + \frac{x \times R \times 1}{100} \right) + \left(x + \frac{x \times R \times 2}{100} \right) + \dots \right]$$

where A = Total amount paid

x = value of each installment

Example: A scooter is sold by an automobile agency for Rs. 19200 cash or for Rs. 4800 cash down payment together with five equal monthly instalments. If the rate of interest charged by the company is 12% per annum, then find each instalment?

Sol. Balance of the price to be paid through instalments

$$P = 19200 - 4800 = 14400$$

Now, according to the formula,

$$A = \left[x + \left(x + \frac{x \times R \times 1}{100} \right) + \left(x + \frac{x \times R \times 2}{100} \right) + \dots + \left(x + \frac{x \times R \times 4}{100} \right) \right]$$

$$\text{where, } A = P + \frac{P \times n \times R}{100}$$

$$\Rightarrow \left(14400 + \frac{14400 \times 12 \times 5}{100 \times 12} \right) = \left[x + \left(x + \frac{12x}{12 \times 100} \right) + \left(x + \frac{12x \times 2}{12 \times 100} \right) + \dots + \left(x + \frac{12x \times 4}{12 \times 100} \right) \right]$$

$$\Rightarrow 15120 = 5x + \frac{x}{10} \Rightarrow x = \frac{151200}{51} \Rightarrow x = \text{Rs. } 2964.70$$

Compound Interest

If Principal = Rs. P, Time = n years, Rate = r% per annum and interest compounded annually

(i) When interest compounded annually $\Rightarrow$ Amount = $P \left[1 + \frac{r}{100} \right]^n$

(ii) When interest compounded half yearly $\Rightarrow$ Amount = $P \left[1 + \frac{\left(\frac{r}{2}\right)}{100} \right]^{2n}$

(iii) When interest compounded quarterly $\Rightarrow$ Amount = $P \left[1 + \frac{\left(\frac{r}{4}\right)}{100} \right]^{4n}$

(iv) When interest compounded monthly $\Rightarrow$ Amount = $P \left[1 + \frac{\left(\frac{r}{12}\right)}{100} \right]^{12n}$

(v) When time is in fraction of a year, say $3\frac{4}{5}$ years $\Rightarrow$ Amount = $P \left[1 + \frac{r}{100} \right]^3 \left[1 + \frac{\left(\frac{4}{5}r\right)}{100} \right]$

(vi) When rate of interest is $r_1\%$ during first year, $r_2\%$ during 2nd year, $r_3\%$ during 3rd year.

$$\text{Amount} = P \left[1 + \frac{r_1}{100} \right] \left[1 + \frac{r_2}{100} \right] \left[1 + \frac{r_3}{100} \right]$$

Concept 2: $A = P \left(1 + \frac{r}{100} \right)^n$, $CI = A - P = P \left(1 + \frac{r}{100} \right)^n - P$, $CI = P \left[\left(1 + \frac{r}{100} \right)^n - 1 \right]$

Concept 3: A sum of money, placed at compound interest, becomes n times in t years and m times in x years

then, $n^{\frac{1}{t}} = m^{\frac{1}{x}}$

Example: A sum of money at compound interest amounts to thrice itself in 3 years, in how many years will it be 9 times itself?

Sol. $3^{\frac{1}{3}} = 9^{\frac{1}{x}}$, $3^{\frac{1}{3}} = 3^{\frac{2}{x}}$, $\frac{1}{3} = \frac{2}{x}$, $x = 6$ years

Concept 4: Relationship between CI and SI for two years

$$\frac{CI}{SI} = \frac{200 + r}{200}$$

Example: The SI on a certain sum of money for 2 years at 10% per annum is Rs. 400, find CI at the same rate and for the same time.

Sol. $CI = \frac{200}{200} \times 400 = \text{Rs. } 420$

Concept 5: When difference between the CI and SI on a certain sum of money for 2 years at r%, is given by,

$$\text{Difference} = P \left(\frac{r}{100} \right)^2$$

Example: The difference between the CI and SI on a certain sum of money at 5% per annum for 2 years is Rs. 1.50, Find the sum

Sol. $1.5 = \frac{P(5)^2}{100^2}$, $P = \text{Rs. } 600$

Concept 6: The difference between CI and SI on a certain sum for 3 years is given by,

$$\text{Difference} = \frac{Pr^2(300+r)}{100^3}$$

Example: If the difference between CI and SI on a certain sum of money for 3 years at 5% per annum is Rs. 122 find the sum.

Sol. $122 = \frac{P(5)^2(300+5)}{100^3}, P = \text{Rs. } 16000$

Concept 7: If a sum 'A' becomes 'B' in t_1 years at compound rate of interest, then after t_2 years the sum becomes

$$\frac{(B)^{\frac{t_2}{t_1}}}{(A)^{\frac{t_2}{t_1}-1}}$$

Example: Rs. 4800 becomes Rs. 6000 in 4 years at a certain rate of CI. What will be the sum after 12 years.

$$\text{Required Amount} = \frac{(6000)^{\frac{12}{4}}}{(4800)^{\frac{12}{4}-1}} = \frac{(6000)^3}{(4800)^2} = \text{Rs. } 9375$$

Concept 8: If a sum of money Rs. x is divided among 'n' parts in such a manner that when placed at CI, amount obtained in each case remains equal while the rate of interest on each part is $r_1, r_2, r_3, \dots, r_n$ respectively and time period for each part is $t_1, t_2, t_3, \dots, t_n$ respectively. Then the divided parts of sum will be in the ratio of:

$$\frac{1}{\left(1 + \frac{r_1}{100}\right)^{t_1}} : \frac{1}{\left(1 + \frac{r_2}{100}\right)^{t_2}} : \frac{1}{\left(1 + \frac{r_3}{100}\right)^{t_3}} : \dots : \frac{1}{\left(1 + \frac{r_n}{100}\right)^{t_n}}$$

Example: A sum of Rs. 3903 is divided between A and B, so that A's share at the end of the 7 years be equal to B's share at the end of 9 years, CI being 4% find A's share.

$$A's \text{ share} : B's \text{ share} = \frac{1}{\left(1 + \frac{4}{100}\right)^7} : \frac{1}{\left(1 + \frac{4}{100}\right)^9} = 1 : \frac{1}{\left(1 + \frac{4}{100}\right)^2} = 1 : \frac{625}{676} = 676 : 625$$

$$A's \text{ share} = \frac{676}{(676+625)} \times 3903 = \text{Rs. } 2028$$

Types of Questions

1. A sum of money becomes four times in 20 years at SI. Find the rate of interest.
2. A sum becomes two times in 5 years at a certain rate of interest. Find the time in which the same amount will be 8 times at the same rate of interest.

Sol. Given, $T = 20$ years,

Let sum = P

Then, the sum after 20 years = 4P

$$\therefore SI = 4P - P = 3P$$

$$\text{Now, } 3P = \frac{PRT}{100} = \frac{P \times R \times 20}{100}$$

$$\Rightarrow 3 = \frac{20R}{100} = \frac{R}{5}$$

$$\therefore R = 15\%$$

Sol. Let sum = P

Then, for 5 years,

$$SI = 2P - P = P$$

$$SI = \frac{P \times R \times T}{100}$$

$$\therefore P = \frac{P \times R \times 5}{100} = \frac{PR}{20}$$

$$\Rightarrow R = 20\%$$

Again, for another time (T),

$$SI = 8P - P = 7P$$

$$\therefore 7P = \frac{P \times 20 \times T}{100} = \frac{20TP}{100} = \frac{TP}{5}$$

$$\therefore T = 7 \times 5 = 35 \text{ years}$$

3. Ravi lent out a certain sum. He lent $\frac{1}{3}$ part of his sum

at 7% SI, $\frac{1}{4}$ part at 8% SI and remaining part at 10% SI. If Rs. 510 is his annual total annual interest, then find the money lent out.

Sol. Let entire sum = P

Now, according to the question,

$$\frac{1}{3}P \times 7\% + \frac{1}{4}P \times 8\% + \left[1 - \left(\frac{1}{3} + \frac{1}{4}\right)\right] \times P \times 10\% = 510$$

$$\Rightarrow \frac{\frac{1}{3}P \times 7}{100} + \frac{\frac{1}{4}P \times 8}{100} + \frac{\frac{5}{12}P \times 10}{100} = 510$$

$$\Rightarrow \frac{7P}{3} + 2P + \frac{25P}{6} = 510 \times 100$$

$$\therefore P = \frac{510 \times 100 \times 6}{51} = \text{Rs. } 6000$$

4. A sum of Rs. 7700 is lent out in two parts in such a way that the interest on one part at 20% for 5 years is equal to that on another part at 9% for 6 years. Find the two sums.

Sol. Let the first sum be Rs. x. Then, second sum = Rs. (7700 - x)

Now, according to the question,

$$\frac{x \times 20 \times 5}{100} = \frac{(7700 - x) \times 9 \times 6}{100}$$

$$\Rightarrow 50x = (7700 - x) \times 27$$

$$\Rightarrow 50x = 7700 \times 27 - 27x$$

$$\Rightarrow 77x = 7700 \times 27$$

$$\Rightarrow x = \text{Rs. } 2700$$

$$\therefore \text{The second part} = (7700 - x) = 7700 - 2700 = \text{Rs. } 5000$$

5. What annual payment will discharge a debt of Rs. 848 in 8 years at 8% per annum?

Sol. Here, P = Rs. 848, T = 8 years, R = 8%

According to the formula,

$$\text{Annual payment} = \frac{100P}{100T + \frac{RT(T-1)}{2}}$$

$$= \frac{100 \times 848}{100 \times 8 + \frac{8 \times 8(8-1)}{2}}$$

$$= \frac{848 \times 100}{800 + 32 \times 7} = \frac{84800}{1024} = \text{Rs. } 82.8125$$

6. Raviraj invested an amount of Rs. 10000 at compound interest rate of 10% per annum for a period of three years. How much amount will Raviraj get after 3 years?

Sol. Amount = $P \left[1 + \frac{r}{100}\right]^n$

where P → Principal, r → Rate, n → time

$$\text{Amount} = 10000 \left(1 + \frac{10}{100}\right)^3 = \text{Rs. } 13310$$

7. Seema invested an amount of Rs. 16000 for two years at compound interest and received an amount of Rs. 17640 on maturity. What is the rate of interest?

Sol. $\frac{A}{P} = \left(1 + \frac{r}{100}\right)^n$

$$\frac{17640}{16000} = \left(1 + \frac{r}{100}\right)^2 = \frac{1764}{1600}$$

$$\left(\frac{42}{40}\right)^2 = \left(1 + \frac{r}{100}\right)^2, \quad r = 5\%$$

8. Rs. 7500 is borrowed at CI at the rate of 4% per annum. What will be the amount to be paid after 1 year, if interest is compounded half-yearly?

Sol. $A = P \left[1 + \frac{r}{100}\right]^{2t} = 7500 \left[1 + \frac{4/2}{100}\right]^{2 \times 1} = \text{Rs. } 7803$

9. Rs. 7500 is borrowed at CI at the rate of 4% per annum. What will be the amount to be paid after 6 months, if interest is compounded quarterly?

Sol. $A = P \left[1 + \frac{r/4}{100}\right]^{4t} = 7500 \left[1 + \frac{4/4}{100}\right]^{4 \times \frac{6}{12}} = \text{Rs. } 7650.75$

10. Find the compound interest on Rs. 2000 at 5% per annum, compounded yearly for $2\frac{1}{2}$ years?

Sol. $A = P \left[1 + \frac{r}{100}\right]^2 \left[1 + \frac{r/2}{100}\right]$

$$= 2000 \left[1 + \frac{5}{100}\right]^2 \left[1 + \frac{5}{200}\right] = 2260.125$$

$$\text{CI} = 2260.125 - 2000 = \text{Rs. } 260.125$$

11. An amount of money grows upto Rs. 4840 in 2 years and upto Rs. 5324 in 3 years on compound interest. Find the rate per cent.

Sol. P + CI of 3 years = Rs. 5324
P + CI of 2 years = Rs. 4840
CI for 3rd year = 5324 - 4840 = Rs. 484

$$\text{Required } r\% = \frac{484 \times 100}{4840} = 10\%$$

12. Find the ratio of CI to SI on a certain sum at 10% per annum for 2 years?

Sol. $\frac{\text{CI}}{\text{SI}} = \frac{200 + r}{200} = \frac{210}{200} = 21 : 20$

13. Rs. 1600 becomes Rs. 2000 in 2 years at a certain rate of compound interest. What will be the sum after 4 years?

Sol. If a sum 'A' becomes 'B' in t_1 years at CI, then after t_2

years, the sum becomes $\frac{(B)^{\frac{t_2}{t_1}}}{(A)^{\frac{t_2}{t_1}-1}}$

$$\text{Required Amount} = \frac{(2000)^{\frac{4}{2}}}{(1600)^{\frac{4}{2}-1}} = \frac{(2000)^2}{(1600)^1} = \text{Rs. } 2500$$

14. What sum of money at compound interest will amount to Rs. 2249.52 in 3 years, if the rate of interest is 3% for the first year, 4% for the second year and 5% for the third year?

Sol. $A = P \left(1 + \frac{r_1}{100}\right) \left(1 + \frac{r_2}{100}\right) \left(1 + \frac{r_3}{100}\right)$

$$2249.52 = P \left(1 + \frac{3}{100}\right) \left(1 + \frac{4}{100}\right) \left(1 + \frac{5}{100}\right)$$

$$P = \text{Rs. } 2000$$

15. Find the difference between CI and SI on Rs. 8000 for 3 years at 2.5% pa.

Sol. Difference = $\frac{\text{Pr}^2(300+r)}{100^3}$ [only possible for 3 years]

$$= \frac{8000(2.5)^2(300+2.5)}{100^3} = \text{Rs. } 15.125$$

Foundation

Questions

1. What is S.I. of Rs. 800 on 5% per annum for 3 years?
(a) Rs. 120 (b) Rs. 140
(c) Rs. 210 (d) Rs. 230
2. How many years will it take for amount of Rs. 600 to yield Rs. 120 as interest at 10% per annum of S.I.?
(a) 1 year (b) 3 year
(c) 2 year (d) 4 year
3. A sum of Rs. 15000 amount gave Rs. 4500 as interest in 5 year. What is rate of interest?
(a) 3% (b) 4%
(c) 5% (d) 6%
4. A sum of money becomes Rs. 1100 in 2 years and Rs. 1400 in 6 years at S.I. Find the rate percent.
(a) $7\frac{17}{19}$ (b) $1\frac{7}{13}$
(c) $2\frac{4}{3}$ (d) $5\frac{2}{3}$
5. What would be the C.I. on Rs. 17500 at the rate of 12 p.c.p.a. after 2 years?
(a) Rs. 4442 (b) Rs. 4452
(c) Rs. 4462 (d) Rs. 4482
6. What would be the C.I. obtained on an amount of Rs. 12000 at the rate of 9 p.c.p.a. for 3 years?
(a) Rs. 3840 (b) Rs. 3740.75
(c) Rs. 3540 (d) Rs. 3640
7. What would be the C.I. obtained on an amount of Rs. 4800 at the rate of 5 p.c.p.a. for 3 years?
(a) Rs. 448.7 (b) Rs. 817.8
(c) Rs. 623.5 (d) Rs. 756.5
8. What would be the C.I. obtained on an amount of 12500 at the rate of 12 p.c.p.a. after 2 years?
(a) Rs. 3180 (b) Rs. 3360
(c) Rs. 3540 (d) Rs. 3720
9. The difference between simple and compound interest on sum of 10000 is 64 for 2 years. Find the rate of interest.
(a) 8 (b) 64
(c) 4 (d) 2
10. The difference between simple and compound interest compounded annually on a certain sum of money for 2 years at 4% per annum is Rs. 1. The sum (in Rs.) is:
(a) 625 (b) 630
(c) 640 (d) 650
11. Find the compound interest on Rs. 12500 at 8% per annum for 9 months compounded quarterly.
(a) Rs. 1020 (b) Rs. 1428
(c) Rs. 765 (d) Rs. 550
12. Find the compound Interest on Rs. 32000 at 20% per annum for 1 year, compounded half yearly.
(a) Rs. 6320 (b) Rs. 6720
(c) Rs. 6400 (d) Rs. 6500

13. The difference between C.I. & S.I. on Rs. 700 in 2 years at 5% per annum is:
 (a) Rs. 10 (b) Rs. 5
 (c) Rs. 1.75 (d) Rs. 2.5
14. The difference between the compound and simple interest on a sum of money for 2 years at $6\frac{1}{4}\%$ per annum is Rs. 10. The sum is
 (a) Rs. 2000 (b) Rs. 2200
 (c) Rs. 2560 (d) Rs. 2600
15. If a sum of money doubles itself in 8 years at S.I. then the rate of interest per annum is:
 (a) 10.5 (b) 12.5
 (c) 11.5 (d) 13.5
16. What sum lent at 10% per year on S.I. will amount to Rs. 450 in 2 years?
 (a) Rs. 350 (b) Rs. 375
 (c) Rs. 240 (d) Rs. 280
17. Nutan invest 22400 on S.I. at rate 12 p.c.p.a. How much amount she will get after seven year?
 (a) 41,116 (b) 41,216
 (c) 42,116 (d) 42, 216
18. What time taken by sum of Rs. 7000 to became 10500 at the rate of 5% per annum?
 (a) 8 years (b) 10 years
 (c) 5 years (d) 15 years
19. A sum fetched a total simple interest of Rs. 4016.25 at the rate of 9% p.a. in 5 years. What is the sum?
 (a) Rs. 4462.50 (b) Rs. 8032.50
 (c) Rs. 8900 (d) Rs. 8925
20. If the difference between the C.I. compounded half yearly and simple interest on a sum at 10% per annum for one year is Rs. 25, the sum is:
 (a) Rs. 9000 (b) Rs. 9500
 (c) Rs. 10,000 (d) Rs. 10,500
21. The difference between compound and simple interest at a certain rate on Rs. 2000 at the end of two years is Rs. 12.8. The rate of interest per annum is
 (a) 6.8% (b) 8%
 (c) 12.8% (d) 16.8%
22. If a sum of money at simple interest doubles itself in 6 years, it will become four times in:
 (a) 14 years (b) 12 years
 (c) 18 years (d) 16 years
23. A sum of money doubles in 3 years at compound interest, compounded annually. It will become 4 times of itself in:
 (a) 12 years (b) 9 years
 (c) 8 years (d) 6 years
24. If the amount is $6\frac{1}{4}$ times the sum after 2 years at compound interest, the rate of interest per annum is:
 (a) 130% (b) 140%
 (c) 125% (d) 150%
25. Amit invested an amount of Rs. 25000 in fixed deposit @ C.I. 8% per annum for 2 years. What amount Amit will get?
 (a) Rs. 28240 (b) Rs. 28540
 (c) Rs. 29240 (d) Rs. 29160
26. Find the ratio of S.I. & C.I. on a certain sum of money at 5% per annum for 2 years
 (a) 50 : 51 (b) 40 : 41
 (c) 30 : 31 (d) 45 : 46
27. A sum of money invested at C.I. accrued to Rs. 800 in 3 years and to Rs. 840 in 4 years. The rate of interest per annum is.
 (a) $2\frac{1}{2}\%$ (b) 4%
 (c) 5% (d) $6\frac{2}{3}\%$
28. What is difference between C.I. and S.I. of Rs. 12000 on 5% per annum for 2 years?
 (a) 35 (b) 25
 (c) 30 (d) 40
29. Find the difference between C.I. & S.I. on Rs. 15000 at 3% per annum for 3 years?
 (a) 40.91 (b) 73.91
 (c) 39.91 (d) 30
30. Find the difference between C.I. & S.I. on Rs. 13000 at 4% per annum for 3 years?
 (a) Rs. 63.23 (b) Rs. 73.25
 (c) Rs. 68.25 (d) Rs. 70.29
31. A certain sum of money gives Rs. 500 at 5% S.I. for 1 year. What is the C.I. at the same rate at the end of 2 years?
 (a) Rs. 1050 (b) Rs. 1015
 (c) Rs. 1035 (d) Rs. 1025
32. The difference between the S.I. and C.I. obtained on principal amount at 5 p.c.p.a. after 2 years is Rs. 35. What is principal amount.
 (a) Rs. 15000 (b) Rs. 10000
 (c) Rs. 14000 (d) Rs. 13000
33. The C.I. on Rs. 10000 at 20% per annum at the end of 1 year 6 month, if the interest is calculated half yearly will be
 (a) 4340 (b) 5320
 (c) 3310 (d) 4590
34. The S.I. is 7200 on 12 p.c.p.a. for 6 years on a sum. What is C.I. on 5 p.c.p.a for 2 years?
 (a) Rs. 1020 (b) Rs. 1055
 (c) Rs. 1050 (d) Rs. 1025

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35. If the difference between S.I. and C.I. is Rs. 25 on 10000 for 2 years then what is rate of interest?
 (a) 6 (b) 4
 (c) 5 (d) 3
36. If the ratio of CI & SI interest for the same principal and rate of interest for 2 years is 1.2, find the rate of interest.
 (a) 300% (b) 15%
 (c) 40% (d) 3%
37. What is CI on Rs. 7500 for 4 years if the rate of interest is 10% p.a. for the first 2 year and 20% p.a. for the next 2 year?
 (a) Rs. 5273 (b) Rs. 5568
 (c) Rs. 5724 (d) Rs. 5325
38. What will be CI on an amount of Rs. 10000 at the rate of 20 p.c.p.a. in two years if the interest rate is compounded half yearly?
 (a) Rs. 4525 (b) Rs. 4232
 (c) Rs. 4641 (d) Rs. 4461
39. Find the difference between SI and CI at 12% for 2 year on Rs. 10000.
 (a) Rs. 169 (b) Rs. 12
 (c) Rs. 144 (d) Rs. 24
40. Find the difference between SI and CI at 14% for 2 years on Rs. 10000.
 (a) Rs. 144 (b) Rs. 169
 (c) Rs. 124 (d) Rs. 196
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Moderate

1. Prabhat took a certain amount as a loan from bank at the rate of 8% Simple interest per annum and gave the same amount to Ashish as a loan at the rate of 12% p.a. on S.I. If at the end of 12 yrs, he made a profit of Rs. 320 in the deal, What was the original amount?
 (a) Rs. 500 (b) Rs. 600
 (c) Rs. 666.67 (d) Rs. 750.27
2. What annual instalment will discharge a debt of Rs. 1092 due in 3 years at 12% Simple interest?
 (a) Rs. 250 (b) Rs. 275
 (c) Rs. 300 (d) Rs. 325
3. A Man invests a certain sum of money at 6% p.a. Simple interest and another sum at 7% p.a. Simple interest. His income from interest after 2 years was Rs. 354. One fourth of the first sum is equal to one fifth of the second sum. Find the total sum he invested?
 (a) Rs. 1500 (b) Rs. 1200
 (c) Rs. 2700 (d) Rs. 5400
4. Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B?
 (a) Rs. 6400 (b) Rs. 6500
 (c) Rs. 7200 (d) Rs. 7500
5. An automobile financier claims to be lending money at simple interest, but he includes the interest every six months for calculating the principal. If he is charging an interest of 10%, the effective rate of interest becomes:
 (a) 10% (b) 10.25%
 (c) 10.5% (d) 15%
6. A sum of Rs. 725 is lent in the beginning of a year at a certain rate of interest. After 8 months, a sum of Rs. 362.50 more is lent but at the rate twice the former. At the end of the year, Rs. 33.50 is earned as interest from both the loans. What was the original rate of interest?
 (a) 3.6% (b) 4.5%
 (c) 3.46% (d) 6%
7. Aditya invest Rs. 50,000 in a fixed deposit at 10% C.I. for 2 years. At the end of 2 years she put the money in another deposit at 12% S.I. for 3 years. What was the final value of the initial Investment?
 (a) Rs. 80000 (b) Rs. 81280
 (c) Rs. 82280 (d) Rs. 83280
8. There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000 after 3 years at the same rate?
 (a) Rs. 2160 (b) Rs. 3120
 (c) Rs. 3972 (d) Rs. 6240
9. A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits Rs. 1600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is:
 (a) Rs. 120 (b) Rs. 121
 (c) Rs. 122 (d) Rs. 123
10. Aditya invested certain amount in two different schemes A and B. Scheme A offer S.I @ 12 p.c.p.a. and scheme B offer C.I. @ 10 p.c.p.a. Interest accrued on the amount invested in scheme A in 2 years was Rs. 3600 and the total amount invested was Rs. 35000. What was the interest accrued on the amount invested in scheme B?
 (a) Rs. 4500 (b) Rs. 4200
 (c) Rs. 4000 (d) Cannot be determine
11. The population of a town in 2009 was 125000. It increase 10% per year. What is the population after 3 years?
 (a) 166375 (b) 170000
 (c) 125000 (d) 10000
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12. A sum of Rs. 1500 amounts to Rs. 3000 in five years at a certain rate of simple interest. If the rate of interest is increased by 1% the same sum in the same time would amount to:
- (a) Rs. 3,288 (b) Rs. 3,312
(c) Rs. 3,340 (d) Rs. 3075
13. Abhinav invested a certain amount at the rate of 8 p.c.p.a. for 5 years and obtained a SI of Rs. 3800. Had he invested the same amount at the same interest for 2 years, how much amount would she have obtained as CI at the end of 2 years?
- (a) Rs. 1520 (b) Rs. 1550.5
(c) Rs. 1550 (d) Rs. 1580.8
14. Nutan borrowed some money at rate of 4 p.c.p.a. for the first three year, at the rate of 8 p.c.p.a. for the next 2 years and at the rate of 9 p.c.p.a. for the period beyond 5 years. If she pay a total SI of Rs. 19550 at the end of 7 years. How much money did she borrow?
- (a) Rs. 41500 (b) Rs. 42500
(c) Rs. 39500 (d) Rs. 40500
15. Girish invested some money in a bank. After eight years his amount became triple. How many times of the original sum will the money become in 20 years at Simple Interest?
- (a) 5 times (b) 6 times
(c) 7 times (d) 8 times
16. Ram divided 2189 in three parts such that interest on them after 1, 2 and 3 years respectively be equal. The rate of SI is 4% per annum in all cases. The smallest part is
- (a) Rs. 702 (b) Rs. 398
(c) Rs. 425 (d) Rs. 756
17. Ravi invested a total amount of Rs. 65000 in three different schemes A, B, C with rate of interest 12 p.c.p.a., 16 p.c.p.a. and 18 p.c.p.a. respectively and earned a total interest of Rs. 10,180 in one year. If the amount invested in A is 72% of the amount invested in C then what was the amount invested in scheme B.
- (a) Rs. 15000 (b) Rs. 25000
(c) Rs. 22000 (d) Rs. 18000
18. The population of Mumbai increase 5% per annum. It's population was 100000 in 2004. What was it's population after 3 year?
- (a) 115760 (b) 123575
(c) 132557 (d) 11000
19. Which of the statement given below are sufficient to answer the following question.
What is the rate of interest p.c.p.a. on the amount of Rs. 15000 taking loan?
- (1) The difference between C.I. & S.I. is Rs. 170
(2) The S.I. of 2 years is Rs. 2500
- (a) Only I (b) Only II
(c) Either I or II (d) neither I or II
20. Which of the statement given below are sufficient to answer the following question.
What is the rate of interest per annum on an investment of Rs. 800?
- (1) The income from S.I. at the end of 3 years at the same rate of interest is 19
(2) C.I. is 15.76 more than S.I. on same sum & same rate in 2 years
- (a) Only I (b) Only II
(c) Either I or II (d) neither I nor II
21. Which information given below is sufficient to know the amount if the difference between the C.I. & S.I. for 2 years is 18?
- (1) The rate is same at which an amount of Rs. 1000 become 1120 for 2 years in S.I.
(2) The principal given is Rs. 2000.
- (a) Only I (b) Only II
(c) Both I or II (d) Neither I nor II
22. A sum of Rs. 1521 is lent out in two parts in such a way that the simple interest on one part at 10% for 5 yr is equal to that of another part at 8% for 10 yr. What will be the two parts of sum?
- (a) Rs. 926 and Rs. 595 (b) Rs. 906 and Rs. 615
(c) Rs. 916 and Rs. 605 (d) Rs. 936 and Rs. 585
23. Rashmi lent Rs. 600 to Geeta for 2 yr and Rs. 150 to Seeta for 4 yr at same rate of interest and received altogether Rs. 80 as simple interest from both. Find the rate of interest.
- (a) $3\frac{4}{9}\%$ (b) $2\frac{4}{9}\%$
(c) $5\frac{4}{9}\%$ (d) $4\frac{4}{9}\%$
24. Hemant makes a fixed deposit of Rs. 20000 in Bank of India for a period of 3 yr. If the rate of interest be 13% SI per annum charged half-yearly, what amount will he get after 42 months?
- (a) Rs. 27800 (b) Rs. 28100
(c) Rs. 29100 (d) Rs. 30000
25. Gaurav borrowed Rs. 800 at 6% and Naresh borrowed Rs. 600 at 10%. After how much time, will they both have equal debts?
- (a) $15\frac{1}{3}$ yr (b) $14\frac{1}{2}$ yr
(c) $18\frac{1}{3}$ yr (d) $16\frac{2}{3}$ yr
26. What annual payment will discharge a debt of Rs. 1092 due in 2 yr at 12% simple interest?
- (a) Rs. 725 (b) Rs. 325
(c) Rs. 515 (d) Rs. 900
-

27. The annual payment of Rs. 160 in 5 yr at 5% per annum simple interest will discharge a debt of
(a) Rs. 980 (b) Rs. 880
(c) Rs. 440 (d) Rs. 220
28. The population of a city increase at the rate of 5% pa. If the present population of the city is 370440, then what was its population 3 yr ago?
(a) 3 lakh (b) 3.2 lakh
(c) 3.4 lakh (d) 3.6 lakh
29. The population of a particular area of a city is 5000. It increases by 10% in 1st yr. It decreases by 20% in the 2nd yr because of some reason. In the 3rd yr, the population increases by 30%. What will be the population of area at the end of 3 years?
(a) 5120 (b) 5300
(c) 5720 (d) 5620
30. The cost price of a car is Rs. 400000. If its price decreases by 10% every year, then what will be the cost of car after 3 years?
(a) Rs. 3,00,000 (b) Rs. 2,91,700
(c) Rs. 2,91,600 (d) Rs. 2,50,000
31. Suneeta borrowed certain sum from Reena for 2 yr at simple interest. Suneeta lent this sum to Venu at the same rate for 2 yr at compound interest. At the end of 2 yr, she received Rs. 110 as compound interest but paid Rs. 100 as simple interest. Find the sum and the rate of interest?
(a) Rs. 250, rate 10% pa (b) Rs. 250, rate 20% pa
(c) Rs. 250, rate 25% pa (d) Rs. 250, rate 30% pa
32. Find the least number of complete years in which a sum of money put out at 20% compound interest will be more than double?
(a) 3 yr (b) 4 yr
(c) 5 yr (d) 8 yr
33. The population of a country is 10 crore and it is the possibility that the population will become 13.31 crore in 3 years. What will be the annual rate per cent of this growth?
(a) 8% (b) 12.7%
(c) 10% (d) 15%
34. Income of Hemant was Rs. 4000. In the first 2 years, his income decreased by 10% and 5% respectively but in the third year, the income increased by 15%. What was his income at the end of third year?
(a) Rs. 3933 (b) Rs. 4000
(c) Rs. 3500 (d) Rs. 3540
35. A man borrows Rs. 5100 to be paid back with compound interest at the rate of 4% pa by the end of 2 years in two equal yearly instalments. How much will be each instalment?
(a) Rs. 2704 (b) Rs. 2800
(c) Rs. 3000 (d) Rs. 2500
36. Divide Rs. 2602 between X and Y, so that the amount of X after 7 yr is equal to the amount of Y after 9 yr, the interest being compounded at 4% pa.
(a) Rs. 1352, Rs. 1250 (b) Rs. 1400, Rs. 1350
(c) Rs. 1215, Rs. 1300 (d) Rs. 1500, Rs. 1450
37. If Rs. 3000 amounts to Rs. 4320 at compound interest in a certain time, then Rs. 3000 amount to what in half of the time?
(a) Rs. 3400 (b) Rs. 3600
(c) Rs. 38000 (d) Rs. 3520
38. A sum of money lent at compound interest for 2 yr at 20% pa would fetch Rs. 964 more, if the interest was payable half-yearly than if it was payable annually. What is the sum?
(a) Rs. 40000 (b) Rs. 60000
(c) Rs. 90000 (d) Rs. 500000
39. A sum of Rs. 8448 is to be divided between X and Y who are respectively 18 and 19 yr old, in such a way that if their shares be invested at 6.25% per annum at compound interest, they will receive equal amounts on attaining the age of 21 yr. The present share of X is
(a) Rs. 4225 (b) Rs. 4352
(c) Rs. 4096 (d) Rs. 4000
40. During the first year, the population of a village is increased by 5% and in second year it is diminished by 5%. At the end of the second year, its population was 47880. What was the population at the beginning of the first year?
(a) 45500 (b) 48000
(c) 43500 (d) 53000

Difficult

1. According to a census report, the population growth rate of Faridabad is going to be an increasing A.P. with first year's rate as 5% and common difference as 5%, but simultaneously the migration rate is an increasing G.P. with first term as 1% and common ratio of 2. If population on 31 December 2010 is 1 million, then find in which year will Faridabad witness its first fall in population?
(a) 2016 (b) 2014
(c) 2013 (d) 2019
2. Monica deposited a total of Rs. 10500 with a bank in two different deposit schemes at 10% p.a., interest being compounded annually. As per the schemes, she gets the same amount after 2 years on the first deposit as she gets after 3 years on the second deposit. How much money did she deposit for 3 years?

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- (a) Rs. 4500 (b) Rs. 5000
(c) Rs. 6500 (d) Rs. 7200
3. A sum of money invested at simple interest for two years at 5% p.a. amounted to Rs. 2750. The rate at which the sum should have been invested to get Rs. 300 as interest after two years would be?
(a) 6% (b) 6.9%
(c) 7.6% (d) 8.1%
4. A sum of Rs. 1000 after 3 years at compound interest becomes a certain amount that is equal to the amount that is the result of a 3 year depreciation from Rs. 1728. Find the difference between the rates of C.I. and depreciation? (Given C.I. is 10% p.a.)
(a) 1.7% (b) 2.9%
(c) 4% (d) 7.6%
5. When the rate of interest in a bank is reduced from 5% p.a. to 4.5% p.a., Bani deposited Rs. 4000 in her saving account. She found that the new interest income remained the same as before. Find her original deposit?
(a) Rs. 3600 (b) Rs. 3150
(c) Rs. 2980 (d) Rs. 2768
6. Two equal sums were lent at simple interest for 4 years and for 3 years respectively. The rate of interest in the latter case was 3% higher than that of the former case, but the amount in each was Rs. 1088. Find the rate of interest of the former case?
(a) 7% (b) 9%
(c) 11% (d) 13%
7. Kavita invested $\frac{1}{3}$ rd of her money at 4% p.a., $\frac{1}{4}$ th at 3% p.a. and the rest at 5% p.a. She received an annual interest of Rs. 500 on her money. Find the total money invested.
(a) Rs. 21500 (b) Rs. 16480
(c) Rs. 12000 (d) Rs. 11200
8. Shruti borrowed Rs. 2500 from her two friends. For one loan she paid 8% p.a. and for the other 6% p.a. simple interest. If she paid Rs. 180 as total interest for one year, how much did she borrow at 8% p.a.?
(a) Rs. 1500 (b) Rs. 1700
(c) Rs. 1950 (d) Rs. 2120
9. A sum of Rs. 2540 is lent out in two parts, one at 12% and the other at $12\frac{1}{2}\%$. If the total annual return on the total money invested is Rs. 312.42, find the money lent at 12% p.a.
(a) Rs. 996 (b) Rs. 1016
(c) Rs. 1265 (d) Rs. 1295
10. A part of Rs. 38800 is lent out at 72% for six months. The rest of the amount is lent out at 5% p.a. after one year. The ratio of interest after 3 years from the time when first amount was lent out is 5 : 4. Find the second part that was lent out at 5%.
- (a) Rs. 28800 (b) Rs. 29586
(c) Rs. 31776 (d) Rs. 32846
11. A certain sum of money invested at compound interest becomes 1.44 times of itself in 2 years. If twice this sum were lent at simple interest, in how many years would it double itself?
(a) 2 years (b) 2.5 years
(c) 3 years (d) 5 years
12. Arun gives his wife Asha a birthday gift, giving her each year a number of rupees equal to the number of years of her age. If her birthday falls on August 8, what sum must be placed at simple interest at 7% on January 1, before she is 42, in order to raise the required sum?
(a) Rs. 1000 (b) Rs. 1200
(c) Rs. 1500 (d) Rs. 1900
13. Bindu lends Rs. 10000 in four parts to her friends. If she gets 8% on Rs. 2000, 7.5% on Rs. 4000 and 8.5 % on Rs. 1400, what percent must she get for the remainder, if the average interest is 8.13%.
(a) 6.3% (b) 8.6%
(c) 9% (d) 9.2%
14. A sum of money at compound interest amounts in two years to Rs. 2809, and in three years to Rs. 2977.54. Find the rate of interest and the original sum.
(a) 4%, Rs. 2500 (b) 6%, Rs. 1800
(c) 4%, Rs. 1800 (d) 6%, Rs. 2500
15. Sapna borrowed a certain sum of money from Kavita under the following repayment scheme based on simple interest. 8% p.a. for the initial 2 years, 9.5% p.a. for the next 4 years, 11% p.a. for the next 2 years, 12% p.a. after the first 8 years. Find the amount which a sum of Rs. 9000 taken for 12 years becomes at the end of 12 years.
(a) Rs. 20160 (b) Rs. 22350
(c) Rs. 23470 (d) Rs. 24567
16. If Rs. 1200 amounts to Rs. 1323 in 2 years at compound interest compounded annually, what will Rs. 1600 amount to in 3 years at the same rate?
(a) Rs. 1780.45 (b) Rs. 1852.50
(c) Rs. 1945.55 (d) Rs. 2166.76
17. If the compound interest on a certain sum for 3 years at 5% p.a. exceeds the simple interest on the same sum and for the same time and at the same rate by Rs. 183, find the sum.
(a) Rs. 17560 (b) Rs. 21680
(c) Rs. 24000 (d) Rs. 26780
18. If the compound interest on a certain sum of money for 3 years at 5% p.a. is Rs. 504.40, find the corresponding simple interest.
(a) Rs. 329.24 (b) Rs. 384.78
(c) Rs. 435.45 (d) Rs. 480.00
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19. Find the least number of complete years in which a sum of Rs. 1500 at 25% compound interest will be more than doubled.
- (a) 4 years (b) 5 years
(c) 6 years (d) 7 years
20. If the compound interest on a certain sum of money for 2 years is Rs. 164 whereas the simple interest for 3 years at the same rate is Rs. 240 on the same sum, find the sum.
- (a) Rs. 1200 (b) Rs. 1400
(c) Rs. 1600 (d) Rs. 1800

Previous Year (Memory Based)

1. A sum of money becomes $\frac{7}{6}$ of itself in 3 years at a certain rate of simple interest. The rate of interest per annum is:
- (a) $5\frac{5}{9}\%$ (b) $6\frac{5}{9}\%$
(c) 18% (d) 25%
2. The simple interest on a certain sum at 5% per annum for 3 years and 4 years differ by Rs. 42. The sum is:
- (a) Rs. 210 (b) Rs. 280
(c) Rs. 750 (d) Rs. 840
3. A sum of Rs. 10,000 is lent partly at 8% and remaining at 10% per annum. If the yearly interest on the average is 9.2%, the two parts are:
- (a) Rs. 4000, Rs. 6000 (b) Rs. 4500, Rs. 5500
(c) Rs. 5000, Rs. 5000 (d) Rs. 5500, Rs. 4500
4. The simple interest on a sum of money is $\frac{4}{9}$ of the principal and the number of years is equal to the rate percent per annum. The rate per annum is:
- (a) 5% (b) $6\frac{2}{3}\%$
(c) 6% (d) $7\frac{1}{5}\%$
5. Rs. 500 was invested at 12% per annum simple interest and a certain sum of money invested at 10% per annum simple interest. If the total interest on both the sum after 4 years is Rs. 480, the latter sum of money is:
- (a) Rs. 450 (b) Rs. 750
(c) Rs. 600 (d) Rs. 550
6. A money lender finds that due to reducing the interest rate decrease from 8% to $7\frac{3}{4}\%$, his yearly income diminishes by Rs. 61.50. His capital is
- (a) Rs. 22400 (b) Rs. 23800
(c) Rs. 24600 (d) Rs. 26000
7. If the simple interest on a certain sum of money for 15 months at $7\frac{1}{2}\%$ per annum exceeds the simple interest on the same sum for 8 months at $12\frac{1}{2}\%$ per annum by Rs. 32.50, then the sum of money (in Rs.) is:
- (a) 312 (b) 312.50
(c) 3120 (d) 3120.50
8. Ramesh deposited Rs. 15600 in a fixed deposit at the rate of 10% per annum simple interest. After every second year, he adds his interest earnings to the principal. The interest at the end of fourth year is
- (a) Rs. 1716 (b) Rs. 1560
(c) Rs. 3744 (d) Rs. 1872
9. In how many years will a sum of money double itself at $6\frac{1}{4}\%$ simple interest per annum?
- (a) 24 years (b) 20 years
(c) 16 years (d) 12 years
10. A sum of money was invested at a certain rate of simple interest for 2 years. Had it been invested at 1% higher rate, it would have fetched Rs. 24 more interest. The sum of money is:
- (a) Rs. 1200 (b) Rs. 1050
(c) Rs. 1000 (d) Rs. 9600
11. The population of a village decreases at the rate of 20% per annum. If its population 2 years ago was 10,000, the present population is:
- (a) 4600 (b) 6400
(c) 7600 (d) 6000
12. Rs. 12000 is divided into two parts such that the simple interest on the first part for 3 years at 12% per annum may be equal to the simple interest on the second part for $4\frac{1}{2}$ years at 16% per annum. The ratio of the first part to the second part is:
- (a) 2 : 1 (b) 1 : 2
(c) 2 : 3 (d) 3 : 2
13. If x, y, z, are three sum of money such that y is the simple interest on x and z is the simple interest on y for the same time and at the same rate of interest, then we have:
- (a) $z^2 = xy$ (b) $xyz = 1$
(c) $x^2 = yz$ (d) $y^2 = zx$
14. A sum of Rs. 7,930 is divided into 3 parts and given on loan at 5% simple interest to A, B and C for 2, 3 and 4 years respectively. If the amounts of all three are equal after their respective periods of loan, then A received a loan of

- (a) Rs. 2,800 (b) Rs. 3,050
(c) Rs. 2,750 (d) Rs. 2,760
15. The principal which gives Rs. 1 interest per day at a rate of 5% simple interest per annum is
(a) Rs. 5000 (b) Rs. 355000
(c) Rs. 7300 (d) Rs. 3650
16. A boy aged 12 years is left with Rs. 100,000 which is under a trust. The trustees invest the money at 6% per annum and pay the minor boy a sum of Rs. 2500, for his pocket money at the end of each year. The expenses of trust come out to be Rs. 500 per annum. Find the amount that will be handed over to the minor boy after he attains the age of 18 years.
(a) Rs. 125000 (b) Rs. 118000
(c) Rs. 150000 (d) Rs. 120000
17. Tushar borrowed a sum of Rs. 12,000 at 15% per annum from a money-lender on 13th January, 1987 and return the amount on 8th June, 1987 to clear his debt. Then the amount paid by Tushar to the money-lender to clear his debt was.
(a) Rs. 15,000 (b) Rs. 11,394
(c) Rs. 12,720 (d) Rs. 13,650
18. The compound interest on Rs. 10,000 in 2 years at 4% per annum, the interest being compound half-yearly, is:
(a) Rs. 636.80 (b) Rs. 824.32
(c) Rs. 912.86 (d) Rs. 825.82
19. The sum of money invested at compound interest amounts in 3 years to Rs. 2,400 and in 4 years to Rs. 2,520. The interest rate per annum is:
(a) 5% (b) 6%
(c) 10% (d) 12%
20. On a certain sum of money the compound interest for 2 years is Rs. 282.15 and the simple interest for the same period of time is Rs. 270. The rate of interest per annum is:
(a) 6.07% (b) 10%
(c) 9% (d) 12.15%
21. In what time will Rs. 1000 becomes Rs. 1331 at 10% per annum, if compounded annually?
(a) 3 years (b) $2\frac{1}{2}$ years
(c) 2 years (d) $3\frac{1}{2}$ years
22. A sum of money on compound interest amounts to Rs. 10648 in 3 years and Rs. 9680 in 2 years. The rate of interest per annum is:
(a) 5% (b) 10%
(c) 15% (d) 20%
23. The compound interest on Rs. 16,000 for 9 months at 20% per annum, interest being compounded quarterly, is:
(a) Rs. 2,520 (b) Rs. 2,524
(c) Rs. 2,522 (d) Rs. 2,518
24. The compound interest of Rs. 2000 in 2 years if the rate of interest is 4% per annum for the first year and 3% per annum for the second year, will be:
(a) Rs. 142.40 (b) Rs. 140.40
(c) Rs. 141.40 (d) Rs. 143.40
25. A money-lender borrows money at 4% per annum and pays the interest at the end of the year. He lends it at 6% per annum at compound interest compounded half yearly and receives the interest at the end of the year. In this way, he gains Rs. 104.50 a year. The amount of money he borrows, is:
(a) Rs. 6,000 (b) Rs. 5,000
(c) Rs. 7,500 (d) Rs. 8000
26. A sum of money at compound interest amounts to thrice itself in 3 years. In how many years will it be 9 times itself?
(a) 9 years (b) 27 years
(c) 6 years (d) 3 years
27. The difference between the compound interest and the simple interest on a certain sum at 5% per annum for 2 years is Rs. 1.50. The sum is:
(a) Rs. 600 (b) Rs. 500
(c) Rs. 400 (d) Rs. 300
28. What sum will give Rs. 244 as the difference between simple interest and compound interest at 10% in $1\frac{1}{2}$ years compounded half yearly?
(a) Rs. 40,000 (b) Rs. 36,000
(c) Rs. 32,000 (d) Rs. 28,000
29. An amount of money appreciates to Rs. 7,000 after 4 years and to Rs. 10,000 after 8 years at a certain compound interest compounded annually. The initial amount of money was
(a) Rs. 4700 (b) Rs. 4900
(c) Rs. 4100 (d) Rs. 4300
30. What does Rs. 250 amounts to in 2 years with compound interest at the rate of 4% in the 1st year and 8% in the second year?
(a) Rs. 280 (b) Rs. 280.80
(c) Rs. 468 (d) Rs. 290.80
31. Two years ago, the value of my motorbike was Rs. 62500. If the value depreciates by 4% every year, now its value is:
(a) Rs. 56700 (b) Rs. 57600
(c) Rs. 57500 (d) Rs. 55700

32. A sum of Rs. 210 was taken as a loan. This is to be paid back in two equal installments. If the rate of interest be 10% compounded annually, then the value of each installment is
 (a) Rs. 127 (b) Rs. 121
 (c) Rs. 210 (d) Rs. 225
33. A certain amount of money earns Rs. 540 as simple interest in 3 years. If it earns a compound interest of Rs. 376.20 at the same rate of interest in 2 years, find the amount (in Rupees).
 (a) 1600 (b) 1800
 (c) 2100 (d) 2000
34. In what time will Rs. 64,000 amount to Rs. 68,921 at 5% per annum interest being compounded half yearly?
 (a) $1\frac{1}{2}$ years (b) 2 years
 (c) 3 years (d) $2\frac{1}{2}$ years.
35. A sum of money is paid back in two annual installments of Rs. 17,640 each, allowing 5% compound interest compounded annually. The sum borrowed was
 (a) Rs. 32,800 (b) Rs. 32,400
 (c) Rs. 32,000 (d) Rs. 32,200
36. A sum of money placed at compound interest doubles itself in 5 years. It will amount to eight times itself at the same rate of interest in
 (a) 15 years (b) 10 years
 (c) 12 years (d) 20 years
37. On what sum of money will the difference between simple interest and compound interest for 2 years at 5% per annum be equal to Rs. 63
 (a) Rs. 23200 (b) Rs. 29200
 (c) Rs. 25200 (d) Rs. 31200
38. The amount on Rs. 25000 in 2 years at annual compound interest, if the rates for the successive years be 4% and 5% per annum respectively is:
 (a) Rs. 26800 (b) Rs. 27300
 (c) Rs. 28500 (d) Rs. 30000
39. At what rate of compound interest per annum will a sum of Rs. 1200 become Rs. 1348.32 in 2 years?
 (a) 6.5% (b) 4.5%
 (c) 6% (d) 7.5%
40. The compound interest on 12000 for 9 months at 20% per annum, interest being compounded quarterly is:
 (a) Rs. 1891.50 (b) Rs. 2089.70
 (c) Rs. 2136 (d) Rs. 1750

Foundation

Solutions

1. (a); $SI = \frac{PRT}{100} = \frac{800 \times 5 \times 3}{100} = \text{Rs. } 120$
2. (c); $T = \frac{SI \times 100}{PR} = \frac{120 \times 100}{600 \times 10} = 2 \text{ years}$
3. (d); $R = \frac{SI \times 100}{PT} = \frac{4500 \times 100}{15000 \times 5} = 6\%$
4. (a); Total SI difference = 1400 - 1100 = Rs. 300
 Total Time = 6 - 2 = 4 years
 $SI = \frac{300}{4} = \text{Rs. } 75$
 For 2 year $SI = 75 \times 2 = \text{Rs. } 150$
 So, $P = 1100 - 150 = \text{Rs. } 950$
 $R = \frac{75}{950} \times 100 = 7\frac{17}{19}\%$
5. (b); $CI = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$
- $= 17500 \left[\left(1 + \frac{12}{100} \right)^2 - 1 \right] = \text{Rs. } 4452$
6. (c); $CI = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$
 Here, $P = \text{Rs. } 12000$, $R = 9\% \text{ p.a.}$, $n = 3 \text{ years}$
 $= 12000 \left[\left(\frac{109}{100} \right)^3 - 1 \right] = \text{Rs. } 3540 \text{ (Appx.)}$
7. (d); $CI = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$
 $= 4800 \left[\left(1 + \frac{5}{100} \right)^3 - 1 \right] = \text{Rs. } 756.6$
8. (a); $CI = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$
 $= 12500 \left[\left(1 + \frac{12}{100} \right)^2 - 1 \right] = \text{Rs. } 3180$

9. (a); Difference between CI and SI for 2 years = $\frac{PR^2}{100^2}$

$$64 = \frac{10000 \times R^2}{100 \times 100} \Rightarrow R^2 = 64 \Rightarrow R = 8\%$$

10. (a); Difference between CI and SI for 2 years = $\frac{PR^2}{100^2}$

$$\Rightarrow \frac{P \times 16}{100^2} = 1 \Rightarrow P = \text{Rs. } 625$$

11. (c); Because amount compound quarterly

$$\text{So, Year} = \text{year} \times 4, \text{ Rate} = \frac{\text{rate}}{4}$$

$$\therefore \text{year} = \frac{9}{12} \times 4 = 3 \text{ years, rate} = \frac{8}{4} = 2\%$$

$$\therefore \text{CI} = 12500 \left[\left(1 + \frac{2}{100} \right)^3 - 1 \right] = \text{Rs. } 765$$

12. (b); Because the amount is compounded half-yearly

$$\text{So, year} = \text{year} \times 2, \text{ rate} = \frac{\text{rate}}{2}$$

$$\therefore \text{year} = 2 \text{ years, rate} = 10\%$$

$$\Rightarrow \text{CI} = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$$

$$= 32000 \left[\left(1 + \frac{10}{100} \right)^2 - 1 \right] = \text{Rs. } 6720$$

13. (c); Difference between CI and SI for 2 years = $\frac{PR^2}{100^2}$

$$= \frac{700 \times 5 \times 5}{100 \times 100} = \text{Rs. } 1.75$$

14. (c); Difference between CI and SI for 2 years = $\frac{PR^2}{100^2}$

$$\text{Here, Difference} = \text{Rs. } 10, R = \frac{25}{4} \% \text{ per annum}$$

$$\Rightarrow 10 = P \times \left[\frac{25}{4 \times 100} \right]^2 \Rightarrow P = \text{Rs. } 2560$$

15. (b); Here, amount = 2P (Where P = Principal)

$$\text{Hence, SI} = P, \text{ Rate} = \frac{100 \times \text{SI}}{P \times T} = \frac{P \times 100}{P \times 8} = 12.5\%$$

16. (b); Amount = $P + \frac{P \times R \times T}{100}$

$$450 = P + \frac{P \times 2 \times 10}{100} \Rightarrow P = \text{Rs. } 375$$

17. (b); Amount = $P + \frac{P \times R \times T}{100}$

$$= 22400 + \frac{22400 \times 12 \times 7}{100} = \text{Rs. } 41216$$

18. (b); Amount = $P + \frac{P \times R \times T}{100}$

$$10500 = 7000 + \frac{7000 \times 5 \times T}{100} \Rightarrow T = 10 \text{ years}$$

19. (d); SI = $\frac{PRT}{100} \Rightarrow 4016.25 = \frac{P \times 9 \times 5}{100} \Rightarrow P = \text{Rs. } 8925$

20. (c); Year = 2 years,

$$R = \frac{10}{2} = 5\% \quad [\because \text{CI is compounded half-yearly}]$$

$$\text{Difference between CI and SI for 2 years} = \frac{PR^2}{100^2}$$

$$25 = \frac{P \times 25}{100 \times 100}; P = \text{Rs. } 10000$$

21. (b); Difference between CI and SI for 2 years = $\frac{PR^2}{100^2}$

$$12.8 = \frac{2000 \times R^2}{(100)^2} \Rightarrow R = 8\% \text{ per annum}$$

22. (c); Here, Amount = Principal + SI

Now, since amount is given as 2P, hence SI must be equal to P

$$\Rightarrow P = \frac{P \times R \times 6}{100} \quad [\text{Using SI formula}]$$

$$\Rightarrow R = \frac{50}{3} \% \text{ per annum}$$

Now, Amount = 4P (Need to find)

Hence, SI = 3P

$$3P = \frac{P \times \frac{50}{3} \times T}{100} \Rightarrow 300 = \frac{50}{3} \times T \text{ or } T = 18 \text{ years}$$

23. (d); Let the sum of money be Rs. 100

$$\text{Rs. } 100 \xrightarrow{3 \text{ years}} \text{Rs. } 200 \xrightarrow{3 \text{ years}} \text{Rs. } 400$$

Here, Rs. 100 becomes 4 times i.e. Rs. 400 in total 6 years

24. (d); Let amount be 100

$$\text{Then } 100 \times \frac{25}{4} = \text{Rs. } 625 \quad (\text{Final amount})$$

$$\text{So, } 625 = 100 \left[\left(1 + \frac{r}{100} \right)^2 \right]$$

$$\Rightarrow \frac{25}{10} = \left(\frac{100+r}{100} \right) \Rightarrow \frac{5}{2} = \frac{100+r}{100} \Rightarrow r = 150\%$$

25. (d); Here, Amount = $P \left[\left(1 + \frac{R}{100} \right)^2 \right]$

$$= 25000 \left[\left(1 + \frac{8}{100} \right)^2 \right]$$

$$= 25000 \left[\left(\frac{27}{25} \right)^2 \right] = \text{Rs. } 29160$$

26. (b); Here, $\frac{SI}{CI} = \frac{\frac{P \times R \times T}{100}}{P \left[\left(1 + \frac{R}{100} \right)^2 - 1 \right]}$

$$= \frac{\frac{100 \times 5 \times 2}{100}}{100 \left[\left(1 + \frac{5}{100} \right)^2 - 1 \right]} = \frac{\frac{10}{100}}{\frac{41}{400}} = 40 : 41$$

27. (c); CI becomes Rs. 800 in 3 years and becomes Rs. 840 in 4 years

$$\Rightarrow \text{Hence, CI of 1 year} = \text{Rs. } (840 - 800) = \text{Rs. } 40$$

$$\Rightarrow \text{Rate} = \frac{40 \times 100}{800 \times 1} \quad [\text{Here, } P = \text{Rs. } 800]$$

$$= 5\% \text{ per annum}$$

28. (c); Here, difference between SI and CI for 2 years

$$= \frac{PR^2}{(100)^2} = \frac{12000 \times (5)^2}{(100)^2} = \text{Rs. } 30$$

29. (a); Difference between CI and SI for 3 years

$$\Rightarrow \text{Difference} = \frac{PR^2(300+R)}{(100)^3}$$

$$= \frac{(15000)(3)^2(303)}{(100)^3} = \text{Rs. } 40.91 \text{ (appx.)}$$

30. (a); The difference between CI and SI for 3 years

$$\Rightarrow \text{Difference} = \frac{PR^2(300+R)}{(100)^3}$$

$$= \frac{13000 \times (4)^2(304)}{(100)^3} = \text{Rs. } 63.23 \text{ (appx.)}$$

31. (d); Principal = $\frac{SI \times 100}{\text{Rate} \times \text{Time}}$

$$= \frac{500 \times 100}{5 \times 1} = \text{Rs. } 10000$$

$\Rightarrow$ Now CI at 5% rate

$$= 10000 \left[\left(1 + \frac{5}{100} \right)^2 - 1 \right] = \text{Rs. } 1025$$

32. (c); Difference between CI and SI for 2 years = $\frac{PR^2}{100^2}$

$$\Rightarrow 35 = \frac{P(5)^2}{(100)^2} \Rightarrow P = \text{Rs. } 14000$$

33. (c); Rate = $\frac{20}{2} = 10\%$, Time = $\frac{3}{2} \times 2 = 3$ years

[$\therefore$ CI is calculated half yearly]

$$CI = 10000 \left[\left(1 + \frac{10}{100} \right)^3 - 1 \right]$$

$$= 10000 \left[\left(\frac{11}{10} \right)^3 - 1 \right] = \text{Rs. } 3310$$

34. (d); $P = \frac{SI \times 100}{R \times T} = \frac{7200 \times 100}{12 \times 6} = \text{Rs. } 10000$

$$\Rightarrow CI = P \left[\left(1 + \frac{R}{100} \right)^2 - 1 \right]$$

$$= 10000 \left[\left(1 + \frac{5}{100} \right)^2 - 1 \right] = \text{Rs. } 1025$$

35. (c); Difference between CI and SI for 2 years = $\frac{PR^2}{(100)^2}$

$$\Rightarrow 25 = \frac{(10000)R^2}{(100)^2} \Rightarrow R = 5\% \text{ per annum}$$

$$\begin{aligned}
 36. \text{ (c); Here, CI} &= P \left[\left(1 + \frac{R}{100} \right)^2 - 1 \right] \\
 &= P \left[1 + \frac{R^2}{(100)^2} + \frac{2R}{100} - 1 \right] = \frac{PR}{100} \left[\frac{R}{100} + 2 \right] \\
 \Rightarrow \text{SI} &= \frac{P \times 2 \times R}{100} \Rightarrow \frac{\text{CI}}{\text{SI}} = 1.2 = \frac{\frac{PR}{100} \left[\frac{R}{100} + 2 \right]}{\frac{2PR}{100}}
 \end{aligned}$$

$$\Rightarrow \frac{\text{CI}}{\text{SI}} = \frac{200 + R}{200}, \quad 1.2 = \frac{200 + R}{200}, \quad R = 40\%$$

$$37. \text{ (b); CI} = 7500 \left[\left(1 + \frac{10}{100} \right)^2 \times \left(1 + \frac{20}{100} \right)^2 - 1 \right]$$

[Here, R = 10% per annum for first 2 years and R = 20% per annum for next 2 years]

$$= 7500 \left[\left(\frac{11}{10} \right)^2 \times \left(\frac{12}{10} \right)^2 - 1 \right] = 5568$$

38. (c); Here, the amount is compounded half-yearly

$$\text{Hence, Rate} = \frac{R}{2}\% = 10\% \text{ per annum}$$

$$\text{Time} = 2T = 2 \times 2 = 4 \text{ years}$$

$$\begin{aligned}
 \Rightarrow \text{CI} &= P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right] \\
 &= 10000 \left[\left(1 + \frac{10}{100} \right)^4 - 1 \right] = \text{Rs. } 4641
 \end{aligned}$$

39. (c); Difference between SI and CI for 2 years = $\frac{PR^2}{(100)^2}$

$$\Rightarrow \text{Difference} = \frac{(10000)(12)^2}{(100)^2} = \text{Rs. } 144$$

40. (d); Difference between SI and CI for 2 years = $\frac{Pr^2}{100^2}$

$$\text{Difference} = \frac{10000 \times 14^2}{10000} = \text{Rs. } 196$$

Moderate

1. (c); Let the original amount be Rs. P.
T = 12, R₁ = 8%, R₂ = 12%, Profit = 320

$$\frac{P \times T \times R_2}{100} - \frac{P \times T \times R_1}{100} = 320$$

$$\frac{P \times 12 \times 12}{100} - \frac{P \times 8 \times 12}{100} = 320 \Rightarrow P = \frac{2000}{3}$$

$$P = \text{Rs. } 666.67$$

2. (d); Let each instalment be Rs. A

$$A = \frac{100P}{100T + \frac{RT(T-1)}{2}} = \frac{100 \times 1092}{100 \times 3 + 36} = \text{Rs. } 325$$

3. (c); Let the sums be A and B

$$R_1 = 6, R_2 = 7, T = 2$$

$$\frac{P_1 \times R_1 \times T}{100} + \frac{P_2 \times R_2 \times T}{100} = 354$$

$$\Rightarrow \frac{A \times 6 \times 2}{100} + \frac{B \times 7 \times 2}{100} = 354 \Rightarrow 6A + 7B = 17700$$

also one fourth of the first sum is equal to one fifth of the second sum

$$\frac{A}{4} = \frac{B}{5} = 5A - 4B = 0$$

By solving the equations we get,

$$A = 1200, B = 1500$$

$$\text{Total sum} = \text{Rs. } (1200 + 1500) = \text{Rs. } 2700$$

4. (a); Let 14% amount = Rs. x

$$11\% \text{ amount} = \text{Rs. } (13900 - x)$$

$$\frac{x \times 2 \times 14}{100} + \frac{(13900 - x) \times 11 \times 2}{100} = 3508$$

$$6x = 350800 - 13900 \times 22 \Rightarrow x = 7500$$

$$\text{Scheme B} = 13900 - 7500 = \text{Rs. } 6400$$

5. (b); Let the sum be Rs. 100 and interest is calculated half-yearly.

$$\Rightarrow \text{Then, Rate} = \frac{10}{2} = 5\%, \quad \text{Time} = 2 \text{ years}$$

$$\text{Effective rate of interest} = 5 + 5 + \frac{5 \times 5}{100} = 10.25\%$$

6. (c); Let the original rate = R%

$$\text{New rate} = (2R)\%$$

$$\frac{725 \times R \times 1}{100} + \frac{362.50 \times 2R \times 1}{3 \times 100} = 33.50$$

$$\Rightarrow (2175 + 725) R = 33.50 \times 100 \times 3$$

$$\Rightarrow (2900) R = 10050 \Rightarrow R = 3.46\%$$

7. (c); CI for first 2 years

$$= 50000 \left[\left(1 + \frac{10}{100} \right)^2 - 1 \right] = \text{Rs. } 10500$$

$$\text{Amount} = P + CI = 50000 + 10500 = \text{Rs. } 60500$$

$$SI = \frac{P \times R \times T}{100} = \frac{60500 \times 12 \times 3}{100} = \text{Rs. } 21780$$

$$\text{Total amount} = \text{Rs. } 82280$$

8. (c); Let $P = 100$, $A = 100 \times \frac{60}{100} + 100 = \text{Rs. } 160$

$$I = 160 - 100 = \text{Rs. } 60, R = \frac{60 \times 100}{100 \times 6} = 10\%$$

$$CI = \frac{331}{1000} \times 12000 = \text{Rs. } 3972$$

9. (b); Here, CI is calculated on half-yearly basis
Hence, Amount would be defined as:

$$A = \left[1600 \left(1 + \frac{5}{2 \times 100} \right)^2 + 1600 \left(1 + \frac{5}{2 \times 100} \right) \right]$$

$$\left(\text{Where Rate} = \frac{\text{Rate}}{2} = \frac{5}{2} \% \right)$$

$$A = \text{Rs. } \left[1600 \times \frac{41}{40} \left(\frac{41}{40} + 1 \right) \right] = \text{Rs. } 3321$$

$$CI = A - P = \text{Rs. } (3321 - 3200) = \text{Rs. } 121$$

10. (b); Let the money in first installment = Rs. x
Second installment = Rs. $(35000 - x)$
The principal is defined as:

$$\Rightarrow P = \frac{3600 \times 100}{12 \times 2} = \text{Rs. } 15000$$

(Using SI in scheme A)

$$\text{The money in first installment} = \text{Rs. } x = \text{Rs. } 15000$$

$$\text{Scheme B} = \text{Rs. } (35000 - 15000) = \text{Rs. } 20000$$

$$\Rightarrow CI = P \left[\left(1 + \frac{R}{100} \right)^2 - 1 \right]$$

$$= 20000 \left[\left(1 + \frac{10}{100} \right)^2 - 1 \right] = \text{Rs. } 4200$$

11. (a); Let population P be 100

$$100 \xrightarrow{+10\% \uparrow} 110 \xrightarrow{+10\% \uparrow} 121 \xrightarrow{+10\% \uparrow} 133.1$$

$$\text{For } P = 100, \text{ increase in 3 years} = 133.1$$

$$\text{For } P = 125000, \text{ increase in 3 years}$$

$$= \frac{133.1}{100} \times 125000 = 166375$$

12. (d); Here, difference = $SI = 3000 - 1500 = \text{Rs. } 1500$

$$\Rightarrow \text{Rate} = \frac{1500 \times 100}{1500 \times 5} = 20\%, (P = \text{Rs. } 1500)$$

$$\Rightarrow \text{Now, Rate of interest} = (20 + 1)\% = 21\%$$

$$\Rightarrow \text{Hence, new SI} = \frac{1500 \times 5 \times 21}{100} = \text{Rs. } 1575$$

$$\Rightarrow \text{Amount} = \text{Principal} + SI \\ = \text{Rs. } [1500 + 1575] = \text{Rs. } 3075$$

13. (d); Principal = $\frac{3800 \times 100}{8 \times 5} = \text{Rs. } 9500$

$$\Rightarrow CI = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right] \\ = 9500 \left[\left(1 + \frac{8}{100} \right)^2 - 1 \right] = \text{Rs. } 1580.8$$

14. (b); Total SI = Rs. 19550

$$19550 = \frac{P \times R_1 \times T_1}{100} + \frac{P \times R_2 \times T_2}{100} + \frac{P \times R_3 \times T_3}{100}$$

$$19550 = P \left[\frac{4 \times 3}{100} + \frac{8 \times 2}{100} + \frac{9 \times 2}{100} \right]$$

$$19550 = P \left[\frac{12}{100} + \frac{16}{100} + \frac{18}{100} \right] \Rightarrow P = \text{Rs. } 42500$$

15. (b); Let his money be 100 Rs.

$$\text{After eight year it becomes } 300$$

$$\therefore \text{interest} = 200$$

$$\text{So, Rate} = \frac{200 \times 100}{100 \times 8} = 25\%$$

$$\text{In 20 year, SI} = \frac{100 \times 25 \times 20}{100} = 500$$

$$\text{So, amount} = 500 + 100 = \text{Rs. } 600$$

$$\text{Hence, from Rs. } 100 \text{ to Rs. } 600, \text{ it becomes 6 times}$$

16. (b); Let amount be a, b, c respectively then,

$$\frac{a \times 4 \times 1}{100} = \frac{b \times 4 \times 2}{100} = \frac{c \times 4 \times 3}{100}$$

$$\Rightarrow a = 2b = 3c = 6K (\text{let})$$

$$\therefore a = 6K, b = 3K, c = 2K$$

$$\text{so the part are in ratio } 6:3:2$$

$$\therefore \text{Smallest part} = \frac{2189}{11} \times 2 = 199 \times 2 = \text{Rs. } 398$$

17. (c); Suppose amount invested in scheme A = x

$$\text{Suppose amount invested in scheme B} = y$$

$$\text{Suppose amount invested in scheme C} = z$$

$$\therefore x + y + z = 65000 \quad \dots(i)$$

From question

$$x = 72\% \text{ of } z = \frac{18}{25}z \quad \dots(\text{ii})$$

$$\text{So, } 12x + 16y + 18z = 1018000 \\ \Rightarrow 6x + 8y + 9z = 509000 \quad \dots(\text{iii})$$

From (i), (ii) and (iii) get

$$\frac{43}{25}z + y = 65000 \quad \dots(\text{iv})$$

$$\frac{108}{25}z + 8y + 9z = 509000$$

$$\Rightarrow \frac{333}{25}z + 8y = 509000 \quad \dots(\text{v})$$

$$\text{From (iv) we have } \frac{344}{25}z + 8y = 520000 \quad \dots(\text{vi})$$

$$\therefore (\text{vi}) - (\text{v}) \Rightarrow \frac{11}{25}z = 11000$$

$$\therefore 11z = \text{Rs. } 275000, z = \text{Rs. } 25000$$

$$\therefore y = \text{Rs. } 22000$$

18. (a); Here, Population $P = 100000$

$$100000 \xrightarrow{5\% \uparrow} 105000 \xrightarrow{5\% \uparrow} 110250 \xrightarrow{5\% \uparrow} 115762.5$$

(2004) (2005) (2006) (2007)

Hence Population after 3 years is almost 115760

19. (c); This question can be solved either using statement (1) or (2)

In statement (1), it is simple formula of difference

$$= \frac{PR^2}{(100)^2}$$

In statement (2), it is simple formula of SI

20. (c); This question can be solved either using statement (1) or (2)

In statement (1), we can apply formula of SI in statement (2), we need to determine difference

$$\text{formula of } = \frac{PR^2}{(100)^2}$$

21. (c); We can solve this question using both information together.

Because, in statement I, we can find rate of interest and in statement II, we know the Principal.

22. (d); Given, $T_1 = 5 \text{ yr}$, $R_1 = 10\%$ and $T_2 = 10 \text{ yr}$ and $R_2 = 8\%$

Let the first part = x

Then, second part = $(1521 - x)$

Now, according to the question,

$$\frac{x \times 5 \times 10}{100} = \frac{(1521 - x) \times 10 \times 8}{100}$$

$$\Rightarrow 5x = 12168 - 8x \Rightarrow 13x = 12168, x = \text{Rs. } 936$$

and second part = $1521 - 936 = \text{Rs. } 585$

23. (d); Given, $T_1 = 2 \text{ years}$ and $T_2 = 4 \text{ years}$,
 $P_1 = \text{Rs. } 600$, $P_2 = \text{Rs. } 150$ and $SI_1 + SI_2 = \text{Rs. } 80$
According to the question,

$$\frac{600 \times R \times 2}{100} + \frac{150 \times R \times 4}{100} = 80$$

$$\Rightarrow 120R + 60R = 800 \Rightarrow 180R = 800$$

$$\therefore R = \frac{800}{180} = \frac{80}{18} = \frac{40}{9} = 4\frac{4}{9}\%$$

24. (c); Given, time = 42 months

$$\Rightarrow \frac{42}{12} \text{ years} = 3\frac{1}{2} \text{ years}$$

$$\Rightarrow \frac{7}{2} \times 2 = 7 \text{ half-yr, rate} = \frac{13}{2}\% \text{ half-yearly}$$

$$SI = \frac{20000 \times 13 \times 7}{100 \times 2} = \text{Rs. } 9100$$

$$\therefore \text{Amount (A)} = 20000 + 9100 = \text{Rs. } 29100$$

25. (d); Given, $R_1 = 6\%$, $R_2 = 10\%$

According to the question,

$$800 + \frac{800 \times 6 \times T}{100} = 600 + \frac{600 \times 10 \times T}{100}$$

$$800 + 48T = 600 + 60T \Rightarrow 12T = 200 \Rightarrow 3T = 50$$

$$\therefore T = \frac{50}{3} = 16\frac{2}{3} \text{ years}$$

26. (c); According to the formula,

$$\text{Annual payment} = \frac{100P}{100T + \frac{RT(T-1)}{2}}$$

$$= \frac{1092 \times 100}{100 \times 2 + \frac{24(2-1)}{2}} = \frac{1092 \times 100}{212}$$

$$= \text{Rs. } 515.09 \approx \text{Rs. } 515$$

27. (b); Given, annual payment = Rs. 160

$$R = 5\%, T = 5 \text{ years, debt (P)} = ?$$

According to the formula,

$$\text{Annual payment} = \frac{100P}{100T + \frac{RT(T-1)}{2}}$$

$$\Rightarrow 160 = \frac{100P}{5 \times 100 + \frac{5 \times 4 \times 5}{2}} \Rightarrow 160 = \frac{100P}{550}$$

$$\therefore P = \frac{550 \times 160}{100} = 55 \times 16 = \text{Rs. } 880$$

28. (b); Given that, $P = 370440$, $R = 5\%$ and $n = 3$ years
According to the formula,

$$\text{Population } n \text{ years ago} = \frac{P}{\left(1 + \frac{R}{100}\right)^n}$$

$$\text{Population 3 years ago} = \frac{370440}{\left(1 + \frac{5}{100}\right)^3} = \frac{370440}{\left(\frac{21}{20}\right)^3}$$

$$= \frac{370440 \times 20 \times 20 \times 20}{21 \times 21 \times 21} = \frac{370440 \times 8000}{9261}$$

$$= 40 \times 8000 = 320000$$

29. (c); Population of particular area P of a city = 5000

In 1st year population got increased by 10%

$\therefore$ Population $P_1 = 5000 + 10\%$ of 5000 = 5500

In 2nd year population decreased by 20%

$\therefore$ Population $P_2 = 5500 - 20\%$ of 5500

$$= 5500 - \frac{20}{100} \times 5500 = 4400$$

In 3rd year population increased by 30%

Population $P_3 = 4400 + 30\%$ of 4400

$$= 4400 + \frac{30}{100} \times 4400 = 5720$$

30. (c); Given that cost of a car, $P = \text{Rs. } 400000$
 $\Rightarrow$ Decreasing rate = 10%, year = 3 years

$$\Rightarrow \text{Price of car after 3 years} = P \left(1 - \frac{R}{100}\right)^3$$

$$= 400000 \left(1 - \frac{10}{100}\right)^3 = 400000 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10}$$

$$= \text{Rs. } 291600$$

31. (b); Let the sum borrowed be Rs. P

Rate of interest = $R\%$

CI = Rs. 110, SI = Rs. 100, Time (t) = 2 years

$$\text{CI} - \text{SI} = 110 - 100 = 10$$

$$\Rightarrow \text{Difference} = \frac{PR^2}{(100)^2} \Rightarrow 10 = \frac{PR^2}{(100)^2} \quad \dots(i)$$

$$\Rightarrow \text{Difference} = \frac{\text{SI} \times R}{200}$$

$$10 = \frac{100 \times R}{200}; R = 20\% \quad \dots(ii)$$

Put value (ii) in (i)

$$\Rightarrow 10 = \frac{P(20)^2}{(100)^2}; P = \text{Rs. } 250$$

32. (b); Let the sum be Rs. P

According to given question:

$$\Rightarrow P \left(1 + \frac{R}{100}\right)^n > 2P \Rightarrow P \left(1 + \frac{20}{100}\right)^n > 2P$$

$$\Rightarrow \left(\frac{6}{5}\right)^n > 2 \text{ By hit and trial method, } n > 4 \text{ years}$$

33. (c); Given, $P = 10$ crore
and population after 3 years = 13.31 crore
According to the formula,

$$\text{Population after } n \text{ year} = P \left(1 + \frac{R}{100}\right)^n$$

$$\Rightarrow 13.31 = 10 \left(1 + \frac{R}{100}\right)^3 \Rightarrow \frac{1331}{1000} = \left(1 + \frac{R}{100}\right)^3$$

$$\Rightarrow \left(\frac{11}{10}\right)^3 = \left(1 + \frac{R}{100}\right)^3 \Rightarrow 1 + \frac{R}{100} = \frac{11}{10}$$

$$\Rightarrow \frac{R}{100} = \frac{11}{10} - 1 = \frac{1}{10} \therefore R = 10\%$$

34. (a); Initial income of Hemant, $P = \text{Rs. } 4000$

Let r_1 (rate of decrease) = 10%,

r_2 (rate of decrease) = 5%, r_3 (rate of growth) = 15%

So, Net final

$$\text{Income} = P \left(1 - \frac{r_1}{100}\right) \left(1 - \frac{r_2}{100}\right) \left(1 + \frac{r_3}{100}\right)$$

$$= 4000 \left(1 - \frac{10}{100}\right) \left(1 - \frac{5}{100}\right) \left(1 + \frac{15}{100}\right)$$

$$= 4000 \times \frac{9}{10} \times \frac{19}{20} \times \frac{23}{20} = \text{Rs. } 3933$$

35. (a); Let the installment in each part be Rs. P

According to question:

$$\Rightarrow \frac{P}{\left(1 + \frac{R}{100}\right)} + \frac{P}{\left(1 + \frac{R}{100}\right)^2} = \text{Rs. } 5100$$

$$\Rightarrow \frac{P}{\left(1 + \frac{4}{100}\right)} + \frac{P}{\left(1 + \frac{4}{100}\right)^2} = 5100$$

$$\frac{25P}{26} + \frac{625}{676} P = 5100 \Rightarrow 1275P = 5100 \times 676$$

$$P = \frac{5100 \times 676}{1275} = \text{Rs. } 2704$$

36. (a); Let the first part be Rs. a
Second part Rs. (2602 - a)
According to question:
Amount after 7 years = Amount after 9 years

$$\Rightarrow a \left(1 + \frac{r}{100}\right)^7 = (2602 - a) \left(1 + \frac{r}{100}\right)^9$$

$$\Rightarrow \frac{a}{(2602 - a)} = \left(1 + \frac{4}{100}\right)^2$$

$$\Rightarrow \frac{a}{(2602 - a)} = \frac{26}{25} \times \frac{26}{25} = \frac{676}{625}$$

$$\Rightarrow 625a = 676 (2602 - a)$$

$$\Rightarrow a = \frac{676 \times 2602}{1301} = \text{Rs. } 1352$$

$$\text{Second part} = \text{Rs. } (2602 - a) = \text{Rs. } 1250$$

37. (b); Let rate = R% and time = n year

$$\text{Then, } 4320 = 3000 \left(1 + \frac{R}{100}\right)^n$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^n = \frac{4320}{3000} = 1.44$$

$$\therefore \left(1 + \frac{R}{100}\right)^{\frac{n}{2}} = \sqrt{1.44} = 1.2$$

$$\therefore \text{Required amount for } \frac{n}{2} \text{ year}$$

$$= 3000 \left(1 + \frac{R}{100}\right)^{\frac{n}{2}} = 3000 \times 1.2 = \text{Rs. } 3600$$

38. (a); Let the principal be Rs. P
If CI is compounded half-yearly,

$$\Rightarrow CI = P \left[\left(1 + \frac{10}{100}\right)^4 - 1 \right]$$

$$\left(\text{where } R = \frac{R}{2} \% \text{ p.a. and } T = 2T \text{ years}\right)$$

$$\Rightarrow CI = P \left[\left(\frac{11}{10}\right)^4 - 1 \right] = \frac{4641 P}{10000}$$

If CI is compounded annually,

$$\Rightarrow CI = P \left[\left(1 + \frac{20}{100}\right)^2 - 1 \right] = \frac{11P}{25}$$

According to question:

$$\Rightarrow \frac{4641P}{10000} - \frac{11}{25}P = 964$$

$$\Rightarrow P = \frac{964 \times 10000}{241} = \text{Rs. } 40000$$

39. (c); Let shares of X and Y be Rs. x and Rs. (8448 - x), respectively.

Amount got by X after 3 years = Amount got by Y after 2 years

$$x \left(1 + \frac{6.25}{100}\right)^3 = (8448 - x) \left(1 + \frac{6.25}{100}\right)^2$$

$$\Rightarrow 1 + \frac{6.25}{100} = \frac{8448 - x}{x} \Rightarrow 1 + \frac{1}{16} = \frac{8448 - x}{x}$$

$$\Rightarrow \frac{17}{16} = \frac{8448 - x}{x} \Rightarrow 17x = 135168 - 16x \Rightarrow$$

$$x = \text{Rs. } 4096$$

40. (b); Let the population at beginning of first year be 'a'
So, according to question:

$$\Rightarrow a \left[1 + \frac{5}{100}\right] \times \left[1 - \frac{5}{100}\right] = 47880$$

$$\Rightarrow a \times \left[\frac{21}{20}\right] \times \left[\frac{19}{20}\right] = 47880$$

$$\Rightarrow a = \frac{47880 \times 20 \times 20}{21 \times 19} = \text{Rs. } 48000$$

Difficult

1. (a); Here from 2010 on wards we would determine the population of Faridabad.
Here, we would be creating 2 tables, one for population growth rate and another for decrease due to migration. On comparing both the tables, if the migration rate would become greater than population growth rate, then there would be a fall in population.

Year	Growth rate (A.P. where a = 5%, d = 5%)	Migration rate (G.P. where a = 1%, r = 2%)
2011	5%	1%
2012	10%	2%
2013	15%	4%
2014	20%	8%
2015	25%	16%
2016	30%	32%

From 2016 on wards, Migration rate > Growth rate

2. (b); Let the amount invested by Monica in first scheme be

Rs. x and in another be Rs. $(10500 - x)$

Rate of interest = 10% p.a.

According to CI

The amount in 1st scheme after 2 years

= Rs. $1.21x$

The amount in 2nd scheme after 3 years

= Rs. $1.331(10500 - x)$

Now since, she got same amount from both scheme. Hence,

$$\Rightarrow 1.21x = 1.331(10500 - x)$$

$$\Rightarrow 1.21x + 1.331x = 10500 \times 1.331$$

$$\Rightarrow 2.541x = 13975.5$$

$$x = \frac{13975.5}{2.541} = 5500$$

$\Rightarrow$ Hence, her amount in 3 years scheme is

$$\Rightarrow (10500 - 5500) = \text{Rs. } 5000$$

3. (a); Here, $SI = \frac{P \times R \times T}{100}$

$$\Rightarrow A - P = \frac{P \times 5 \times 2}{100} \Rightarrow 2750 = P \left[1 + \frac{10}{100} \right]$$

$\Rightarrow$ Principal $P = \text{Rs. } 2500$

Now, we need SI for Rs. 300 after 2 years.

$$\Rightarrow SI = 300 = \frac{2500 \times R \times 2}{100} \Rightarrow R = 6\%$$

4. (a); The amount of Rs. 1000 at 10% p.a. after 3 years would becomes:

$$\Rightarrow P \left(1 + \frac{R}{100} \right)^3 = A \Rightarrow 1000 \left(1 + \frac{10}{100} \right)^3 = A$$

$$\Rightarrow A = \text{Rs. } 1331$$

Here, now the principal of Rs. 1728 depreciated at rate of $R\%$ per annum and amounts to Rs. 1331

$$\Rightarrow P \left[\left(1 - \frac{R}{100} \right) \right]^3 = 1331$$

$$\Rightarrow 1728 \left[\left(1 - \frac{R}{100} \right) \right]^3 = 1331$$

$$\Rightarrow \left(1 - \frac{R}{100} \right)^3 = \frac{1331}{1728} \Rightarrow 1 - \frac{R}{100} = \frac{11}{12}$$

$$\Rightarrow \frac{R}{100} = \frac{1}{12} \text{ or } R = 8.33\%$$

$\Rightarrow$ Hence rate difference between CI and depreciation is almost = $(10 - 8.33)\% = 1.7\%$.

5. (a); Here, we assume original deposit of Bani be Rs. x .

$\Rightarrow$ The SI obtained in both cases is same

$$\Rightarrow \frac{x \times 5 \times T}{100} = \frac{4000 \times 4.5 \times T}{100}$$

$$[R = 5\% \text{ p.a.}] \quad [R = 4.5\% \text{ p.a.}]$$

$$\Rightarrow 5x = 4000 \times 4.5 \Rightarrow x = \text{Rs. } 3600$$

6. (b); Here, Amount = Rs. 1088

Let principal invested be Rs. x and Let Rate of interest be $R\%$ in first case and $(R + 3)\%$ in later case.

$$\Rightarrow A = P \left(1 + \frac{RT}{100} \right)$$

$$\Rightarrow 1088 = x \left(1 + \frac{4R}{100} \right) \quad \dots(i)$$

$$\Rightarrow 1088 = x \left(1 + \frac{3(R+3)}{100} \right) \quad \dots(ii)$$

$$\text{Hence, } x \left(1 + \frac{4R}{100} \right) = x \left(1 + \frac{3(R+3)}{100} \right)$$

$$\Rightarrow \frac{4R}{100} = \frac{3R+9}{100} \Rightarrow R = 9\% \text{ p.a.}$$

7. (c); Let the total money invested be Rs. P .

According to question:

$$\Rightarrow 500 = \frac{\left(\frac{P}{3} \times 4 \times 1 \right)}{100} + \frac{\left(\frac{P}{4} \times 3 \times 1 \right)}{100} +$$

$$\frac{\left[\left(P - \left(\frac{P}{3} + \frac{P}{4} \right) \right) \times 5 \times 1 \right]}{100}$$

$$\Rightarrow 500 = \frac{4P}{300} + \frac{3P}{400} + \frac{25P}{1200}$$

$$\Rightarrow 500 = \frac{16P}{1200} + \frac{9P}{1200} + \frac{25P}{1200} \Rightarrow P = \text{Rs. } 12000$$

8. (a); Let Shruti borrowed Rs. x from one of her friends.

Then remaining amount borrowed from another friend be Rs. $(2500 - x)$

$\Rightarrow$ Total SI for 1 year = Rs. 180

$$\Rightarrow 180 = \frac{x \times 8 \times 1}{100} + \frac{(2500 - x) \times 6 \times 1}{100}$$

$$\Rightarrow 180 = \frac{8x}{100} + \frac{6(2500 - x)}{100}$$

$$\Rightarrow 180 = \frac{8x}{100} + \frac{15000 - 6x}{100}$$

$$\Rightarrow 180 = \frac{8x - 6x}{100} + 150$$

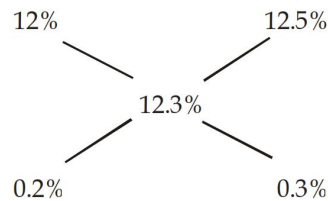
$$\Rightarrow 30 = \frac{2x}{100} \quad x = \text{Rs. } 1500$$

9. (b); Let us evaluate overall rate of interest.

$$\Rightarrow \text{SI} = \text{Rs. } 312.42 = \frac{2540 \times R \times 1}{100}$$

$$\Rightarrow R = \frac{312.42}{25.4} = 12.3\%$$

Now using alligation method, we can determine ratio of sum invested in both parts.



$$\Rightarrow 2 : 3$$

$$\text{Hence, sum lent at } 12\% \text{ p.a.} = \frac{2}{5} \times 2540 = \text{Rs. } 1016$$

10. (a); Let the first part be x. Then the second part be (38800 - x)

$$\text{Now, } \frac{x \times 0.72 \times \frac{1}{2}}{(38800 - x) \times 0.05 \times 2} = \frac{5}{4}$$

$$1.44x = 19400 - 0.5x \Rightarrow x = 10000$$

$$\text{The second part} = 38800 - 10000 = \text{Rs. } 28800$$

11. (d); Let the principal (sum of money) be Rs. 100
 $\Rightarrow$ Amount after 2 years = Rs. 144

$$\Rightarrow 100 \left[\left(1 + \frac{R}{100} \right)^2 \right] = 144 \Rightarrow \left(1 + \frac{R}{100} \right)^2 = \frac{144}{100}$$

$$\Rightarrow 1 + \frac{R}{100} = \frac{12}{10}; R = 20\%.$$

Now, we have to make amount from Rs. 200 to Rs. 400. So, SI would be Rs. 200.

$$\Rightarrow 200 = \frac{200 \times 20 \times T}{100} \Rightarrow T = 5 \text{ years}$$

12. (a); No. of days between 1st January and 8th August (take non-leap year)
 $(30 + 28 + 31 + 30 + 31 + 30 + 31 + 8) = 219$ days

Here, 219 constitutes $\left(\frac{219}{365} \right)$ or $\frac{3}{5}$ th of the year

Thus Arun must invest certain sum for

$\frac{3}{5}$ th of year at 7% to get 42 Rs.

(at 42 years of Asha's age)

$$\Rightarrow \text{SI} = 42 = \frac{P \times 7 \times 3}{100 \times 5} \Rightarrow P = \text{Rs. } 1000$$

13. (c); Here, Let us assume the rate of interest overall be 8.13% (average interest)

$$= \frac{10000 \times 8.13 \times 1}{100} = \text{SI}, \text{SI} = \text{Rs. } 813$$

Now, the interest levied on different parts must be combined to get resultant interest of Rs. 813.

$$\Rightarrow 8\% \text{ of } 2000 + 7.5\% \text{ of } 4000 + 8.5\% \text{ of } 1400 + y\% \text{ of } (10000 - 7400) = 813$$

(where y% = Rate on remainder money)

$$\Rightarrow 8 \times 20 + 75 \times 4 + 8.5 \times 14 + \frac{y}{100} \times 2600 = 813$$

$$\Rightarrow y = 9\%$$

14. (d); The difference in amount of money will determine compound interest of one year

$$\Rightarrow \text{Hence, Rs. } [2977.54 - 2809] = \text{Rs. } 168.54$$

This is the CI levied on Rs. 2809, in one year.

$$\text{Hence, CI} = P \left[1 + \frac{R}{100} - 1 \right]$$

$$\Rightarrow 168.54 = 2809 \left[\frac{R}{100} \right] \Rightarrow \frac{16854}{2809} = R \Rightarrow R = 6\%$$

The amount at CI for 2 years is:

$$\Rightarrow 2809 = P \left[1 + \frac{6}{100} \right]^2 \Rightarrow 2809 = P \left[\frac{106}{100} \times \frac{106}{100} \right]$$

$$\Rightarrow P = \text{Rs. } 2500$$

15. (a); The SI obtained after 12 years would be:

$$\Rightarrow \frac{9000 \times 8 \times 2}{100} + \frac{9000 \times 9.5 \times 4}{100} +$$

$$\frac{9000 \times 11 \times 2}{100} + \frac{9000 \times 12 \times 4}{100}$$

$$\Rightarrow 90 [16 + 38 + 22 + 48] = \text{Rs. } 11160$$

$$\Rightarrow \text{Final amount} = P + \text{SI}$$

$$= \text{Rs. } (9000 + 11160) = \text{Rs. } 20160$$

16. (b); Here, Amount = $P \left[\left(1 + \frac{R}{100} \right)^2 \right]$

$$\Rightarrow 1323 = 1200 \left[\left(1 + \frac{R}{100} \right)^2 \right]$$

$$\Rightarrow \frac{1323}{1200} = \left(1 + \frac{R}{100}\right)^2$$

On solving above equation we get Rate of interest
 $R = 5\%$

Hence, amount of principal = Rs. 1600 in 3 years
 would become:

$$\Rightarrow A = 1600 \left[\left(1 + \frac{5}{100}\right)^3 \right] = 1600 \left[\frac{21}{20} \times \frac{21}{20} \times \frac{21}{20} \right]$$

$$= \frac{1600}{8000} [21]^3 = \frac{1}{5} \times 9261 = \text{Rs. } 1852.50$$

17. (c); According to the question

$$\Rightarrow CI - SI = 183 \quad \dots(i)$$

$$\text{Here, } SI = \frac{P \times 5 \times 3}{100} = \frac{3P}{20} \quad \dots(ii)$$

$$\begin{aligned} \Rightarrow CI &= P \left[\left(1 + \frac{5}{100}\right)^3 - 1 \right] = P \left[\left(\frac{21}{20}\right)^3 - 1 \right] \\ &= P \left[\frac{1261}{8000} \right] \quad \dots(iii) \end{aligned}$$

Putting values of (ii) and (iii) in (i):

$$\Rightarrow \frac{1261}{8000} P - \frac{3P}{20} = 183 \Rightarrow \frac{1261P}{8000} - \frac{1200P}{8000} = 183$$

$$\Rightarrow 61P = 183 \times 8000 \Rightarrow P = \text{Rs. } 24000$$

18. (d); Here, $CI = P \left[\left(1 + \frac{R}{100}\right)^3 - 1 \right]$

$$\Rightarrow 504.40 = P \left[\left(1 + \frac{5}{100}\right)^3 - 1 \right]$$

$$\Rightarrow 504.40 = P \left[\left(\frac{21}{20}\right)^3 - 1 \right] \Rightarrow 504.40 = P \left[\frac{1261}{8000} \right]$$

$$\Rightarrow P(\text{Principal}) = \text{Rs. } 3200$$

$$\Rightarrow SI = \frac{P \times R \times T}{100} = \frac{3200 \times 5 \times 3}{100} = \text{Rs. } 480$$

19. (a); Here, $P = \text{Rs. } 1500$, $R = 25\%$ p.a.

$n = ?$ (No. of years)

Now according to question:

$$\Rightarrow P \left[\left(1 + \frac{R}{100}\right)^n \right] > 2P \Rightarrow \left(1 + \frac{25}{100}\right)^n > 2$$

$$\Rightarrow \left(\frac{5}{4}\right)^n > 2$$

By hit and trial method, we get $\Rightarrow n = 4$ years

20. (c); Here, SI for 3 years = Rs. 240

$$SI \text{ for 1 year} = \text{Rs. } \frac{240}{3}$$

$$SI \text{ for 2 years} = \text{Rs. } \frac{240}{3} \times 2 = \text{Rs. } 160$$

$$\Rightarrow 160 = \frac{P \times R \times 2}{100} \Rightarrow PR = 8000$$

$$\text{Hence, } (164 - 160) = P \left(\frac{R}{100} \right)^2$$

$$\Rightarrow 4 = P \left[\frac{8000}{P \times 100} \right]^2 \Rightarrow 4 = \frac{P \times 8000 \times 8000}{P^2 \times (10000)}$$

$$\Rightarrow P = \text{Rs. } 1600$$

Previous Year (Memory Based)

1. (a); $SI = \frac{7}{6} P - P = \frac{PRT}{100}$

$$\frac{1}{6} P = \frac{PRT}{100} \Rightarrow \frac{1}{6} = \frac{R \times 3}{100}$$

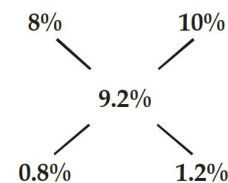
$$R = \frac{100}{18} = 5\frac{5}{9}\%$$

2. (d); $SI_4 - SI_3 = 42$

$$\frac{PR \times 4}{100} - \frac{PR \times 3}{100} = 42$$

$$\frac{PR}{100} = 42 \Rightarrow \frac{P \times 5}{100} = 42$$

3. (a);



$$\frac{A_8}{A_{10}} = \frac{0.8}{1.2} = \frac{2}{3}$$

$$A_8 = \frac{2}{5} \times 10000 = 4000$$

$$A_{10} = 10000 - 4000 = 6000$$

4. (b); Let the interest rate = number of years = x

$$SI = \frac{PRT}{100}, \quad \frac{4}{9}P = \frac{P \times x \times x}{100}$$

$$\frac{400}{9} = x^2 \Rightarrow x = \frac{20}{3}$$

$$\text{rate of interest} = 6\frac{2}{3}\%$$

5. (c); $\frac{500 \times 12 \times 4}{100} + \frac{P \times 10 \times 4}{100} = 480$

$$240 + \frac{2P}{5} = 480 \Rightarrow \frac{2P}{5} = 240 \Rightarrow P = \text{Rs. } 600$$

6. (c); $\left(8 - \frac{31}{4}\right)\% \text{ of } P = 61.5$

$$\frac{1}{4} \times P = 61.5 \Rightarrow P = \text{Rs. } 24600$$

7. (c); $\frac{P \times \frac{15}{2} \times \frac{15}{12}}{100} - \frac{P \times \frac{25}{2} \times \frac{8}{12}}{100} = 32.5$

$$\frac{3P}{32} - \frac{P}{12} = 32.5 \Rightarrow \frac{9P - 8P}{96} = 32.5$$

$$P = 32.5 \times 96 = 3120$$

8. (c); Interest after two year = $\frac{15600 \times 10 \times 2}{100} = \text{Rs. } 3120$

$$\text{Principal becomes after two year} = 15600 + 3120 = 18720$$

$$\text{Interest at the end of fourth year} =$$

$$\frac{18720 \times 2 \times 10}{100} = \text{Rs. } 3744$$

9. (c); $SI = \frac{PRT}{100}$

$$P = \frac{P \times \frac{25}{4} \times T}{100} \Rightarrow T = 16 \text{ years}$$

10. (a); $\frac{P \times (r+1) \times 2}{100} - \frac{P \times r \times 2}{100} = 24$

$$\frac{P \times 2}{100} = 24 \Rightarrow P = \text{Rs. } 1200$$

11. (b); $10000 \frac{-20\%}{1\text{st year}} 8000 \frac{-20\%}{2\text{nd year}} 6400$

12. (a); $P_1 + P_2 = 12000$

$$\frac{P_1 \times 3 \times 12}{100} = \frac{P_2 \times \frac{9}{2} \times 16}{100}$$

$$\frac{P_1}{P_2} = \frac{9 \times 8}{3 \times 12} \Rightarrow \frac{P_1}{P_2} = \frac{2}{1}$$

13. (d); Let rate of interest is r and time is t

$$y = \frac{x \times r \times t}{100}$$

$$\frac{y}{x} = \frac{r \times t}{100} \quad \dots(i)$$

$$z = \frac{y \times r \times t}{100}$$

$$\frac{z}{y} = \frac{r \times t}{100} \quad \dots(ii)$$

from equation (i) and (ii)

$$\frac{y}{x} = \frac{z}{y} \Rightarrow y^2 = xz$$

14. (d); If amount received is equal, then the ratio of distributed amount

$$M_1 : M_2 : M_3 = \frac{1}{100 + r_1 t_1} : \frac{1}{100 + r_2 t_2} : \frac{1}{100 + r_3 t_3}$$

$$= \frac{1}{100 + 5 \times 2} : \frac{1}{100 + 5 \times 3} : \frac{1}{100 + 5 \times 4}$$

$$= \frac{1}{110} : \frac{1}{115} : \frac{1}{120} = \frac{1}{22} : \frac{1}{23} : \frac{1}{24}$$

$$= 276 : 264 : 253$$

$$M_1 = 2760 \Rightarrow M_2 = 2640 \Rightarrow M_3 = 2530$$

$$\text{Amount received by A} = M_1 = \text{Rs. } 2760$$

15. (c); Annual interest = $1 \times 365 = 365$

$$365 = \frac{P \times 5 \times 1}{100}$$

$$P = 365 \times 20 \Rightarrow P = \text{Rs. } 7300$$

16. (b); After 6 year boy will attain age of 18

$$\text{Interest earned from the trust in 6 year}$$

$$= \frac{100000 \times 6 \times 6}{100} = 36000$$

$$\text{Money paid to boy for his pocket money in 6 year}$$

$$= 2500 \times 6 = 15000$$

$$\text{Expenses of trust in 6 year} = 500 \times 6 = 3000$$

$$\text{total amount paid to boy after 6 year}$$

$$= 100000 + 36000 - 15000 - 3000 = \text{Rs. } 118000$$

17. (c); No. of days =

$$\begin{array}{ccccccccc} 18 & + & 28 & + & 31 & + & 30 & + & 31 & + & 8 \\ \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow \\ \text{Jan} & & \text{Feb} & & \text{Mar} & & \text{Apr} & & \text{May} & & \text{June} \\ = 146 \end{array}$$

$$\begin{aligned}\text{Simple interest for 146 days} &= \frac{12000 \times \frac{146}{365} \times 15}{100} \\ &= 720\end{aligned}$$

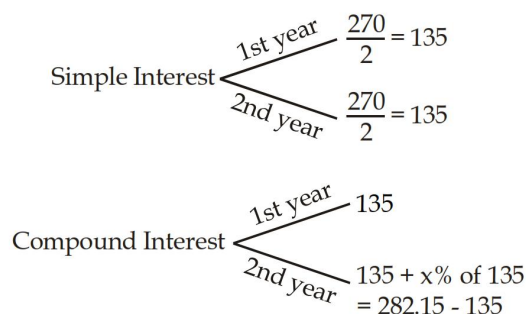
Amount to be paid by Tusar to the money lender
= 12000 + 720 = Rs. 12720

18. (b); $r = \frac{4}{2} = 2\%$

$$\begin{aligned} \text{C.I.} &= 10000 \left(1 + \frac{\frac{4}{2}}{100} \right)^4 - 10000 \\ &= 10000 \left[\left(\frac{102}{100} \right)^4 - 1 \right] \\ &= 10000 \times 0.08243216 = \text{Rs. } 824.3216 \end{aligned}$$

19. (a); rate of interest = $\frac{2520 - 2400}{2400} \times 100$
 $= \frac{120}{2400} \times 100 = 5\%$

20. (c);



$$x\% \text{ of } 135 = 282.15 - 270$$

$$\frac{x}{100} \times 135 = 12.15 \Rightarrow x = \frac{1215}{135} = 9\%$$

21. (a); $1331 = 1000 \left(1 + \frac{10}{100}\right)^t$

$$\left(\frac{11}{10}\right)^3 = \left(\frac{11}{10}\right)^t$$

t = 3 years

22. (b); $9680 = P \left(1 + \frac{r}{100} \right)^2 \quad \dots (i)$

$$10648 = P\left(1 + \frac{r}{100}\right)^3$$

$$10648 = P\left(1 + \frac{r}{100}\right)^2 \left(1 + \frac{r}{100}\right)$$

$$10648 = 9680 \left(1 + \frac{r}{100} \right) \quad \dots [\text{from (i)}]$$

$$\frac{10648}{9680} - 1 = \frac{r}{100}, \quad \frac{968}{9680} = \frac{r}{100}, \quad r = 10\%$$

23. (c); Amount = $16000 \left(1 + \frac{\frac{20}{100}}{3} \right)^9$

$$= 16000 \left(\frac{105}{100} \right)^3 = 16000 \times \frac{9261}{8000}$$

Amount = 18522

$$\text{Compound Interest} = 18522 - 16000 = \text{Rs. } 2522$$

24. (a); Amount = $2000\left(1 + \frac{4}{100}\right)\left(1 + \frac{3}{100}\right)$

$$= 20 \times 104 \times \frac{103}{100} = 104 \times \frac{103}{5} = 2142.4$$

$$CI = 2142.4 - 2000 = \text{Rs. } 142.4$$

25. (b); Let the amount of money borrowed is P.

Amount paid by money lender as a interest at the

$$\text{end of the year} = \frac{P \times 4 \times 1}{100} = \frac{P}{25}$$

Interest received by money lender at the end of the year on the amount which he lend

$$= \left[P \left(1 + \frac{\frac{6}{2}}{100} \right)^2 - P \right] = \left[P \left(\frac{103}{100} \right)^2 - P \right]$$

$$\text{his gain} = \left[P \left(\frac{103}{100} \right)^2 - P \right] - \frac{P}{25}$$

$$104.50 = P \left(\frac{103}{100} \right)^2 - \frac{26}{25}P$$

$$104.50 = P \left[\frac{10609 - 10400}{10000} \right], \quad 1045000 = P \times 209$$

$$P = \frac{1045000}{209} \Rightarrow P = \text{Rs. } 5000$$

26. (c); If a certain amount becomes N times in T years then same amount becomes N^n times in $T \times n$ years

According to question

The sum of money on compound interest becomes 3 times in 3 years then same amount becomes $9 = 3^2$ times in $3 \times 2 = 6$ years

27. (a); Difference between CI and SI = $\frac{PR^2}{100^2}$

$$1.5 = \frac{P \times 5^2}{100^2} \Rightarrow P = \frac{15000}{25} \Rightarrow P = \text{Rs. } 600$$

28. (c); 10% interest rate compounded half yearly for

$1\frac{1}{2}$ year

= 5% interest rate compounded annually for 3 year

Difference between CI and SI for three year

$$= \frac{PR^2}{100^2} \times \frac{300+R}{100}, \quad 244 = \frac{P \times 5^2}{100^2} \times \frac{300+5}{100}$$

$$P = \frac{244 \times 400 \times 100}{305}, \quad P = \text{Rs. } 32000$$

29. (b); Let the amount is P

$$7000 = P \left(1 + \frac{r}{100}\right)^4 \quad \dots(i)$$

$$10000 = P \left(1 + \frac{r}{100}\right)^8$$

$$10000 = P \left(1 + \frac{r}{100}\right)^4 \left(1 + \frac{r}{100}\right)^4$$

$$10000 = 7000 \left(1 + \frac{r}{100}\right)^4$$

$$\left(1 + \frac{r}{100}\right)^4 = \frac{10}{7} \quad \dots(ii)$$

from equation (i) and (ii)

$$7000 = P \times \frac{10}{7} \Rightarrow P = \text{Rs. } 4900$$

30. (b); Amount = $250 \left(1 + \frac{4}{100}\right) \left(1 + \frac{8}{100}\right)$

$$250 \times \frac{104}{100} \times \frac{108}{100} = 280.80$$

31. (b); Value of Bike after depreciation for two years

$$= P \left(1 - \frac{r}{100}\right)^2 = 62500 \times \left(1 - \frac{4}{100}\right)^2$$

$$= 62500 \times \frac{96}{100} \times \frac{96}{100} = 57600$$

32. (b); If x is the amount of each instalment and amount borrowed is P

$$P = x \left[\frac{100}{(100+r)} + \frac{100^2}{(100+r)^2} + \dots + \frac{100^t}{(100+r)^t} \right]$$

$$210 = x \left[\frac{100}{100+10} + \frac{100^2}{(100+10)^2} \right]$$

$$210 = x \times \frac{100}{110} \left[1 + \frac{100}{110} \right]$$

$$210 = \frac{10x}{11} \left[\frac{210}{110} \right], \quad x = 121$$

33. (d); SI for 3 years $\rightarrow 540$

$$\text{SI for 1 year} \rightarrow \frac{540}{3}$$

Let the Rate of interest is r.

$$\text{CI for 2 year} = \underbrace{\frac{180}{100}}_{1\text{st year}} + \underbrace{\frac{180+r\% \text{ of } 180}{100}}_{2\text{nd year}}$$

$$376.20 = 360 + \frac{r}{100} \times 180$$

$$376.20 - 360 = \frac{r}{100} \times 180$$

$$16.20 = \frac{r}{100} \times 180$$

$$\frac{1620}{180} = r \Rightarrow r = 9\%$$

$$\text{SI for 3 year} = \frac{P \times 9 \times 3}{100}$$

$$540 = \frac{P \times 9 \times 3}{100} \Rightarrow P = 2000$$

34. (a); $A = P \left(1 + \frac{r}{100}\right)^t$

t = number of half - year

P = principal amount

$$68921 = 64000 \left(1 + \frac{\frac{5}{2}}{100}\right)^t, \quad \frac{68921}{64000} = \left(1 + \frac{2.5}{100}\right)^t$$

$$\left(\frac{41}{40}\right)^3 = \left(\frac{102.5}{100}\right)^t, \quad \left(\frac{41}{40}\right)^3 = \left(\frac{41}{40}\right)^t$$

$$t = 3 \text{ half - years} = 1\frac{1}{2} \text{ years}$$

35. (a); borrowed amount =

$$x \left[\frac{100}{100+r} + \frac{100^2}{(100+r)^2} + \dots + \frac{100^t}{(100+r)^t} \right]$$

x = amount paid in each installment

r = rate of interest

t = time

$$\text{borrowed amount} = 17640 \left[\frac{100}{100+5} + \frac{(100)^2}{(100+5)^2} \right]$$

$$= 17640 \times \frac{100}{105} \left[1 + \frac{100}{105} \right]$$

$$= 17640 \times \frac{100}{105} \times \frac{205}{105}$$

borrowed amount = Rs. 32800

36. (a); If a certain amount becomes N times in T years then same amount becomes N^n times in $T \times n$ years

According to question,

Sum of money becomes, two times in 5 years then same amount becomes $8 = 2^3$ times in $5 \times 3 = 15$ years

37. (c); Difference between CI and SI for two years = $\frac{PR^2}{100^2}$

$$63 = \frac{P \times 5^2}{100^2} \Rightarrow P = \frac{63 \times 100 \times 100}{5 \times 5}$$

$$\Rightarrow P = \text{Rs. } 25200$$

38. (b); Amount = $P \left(1 + \frac{r_1}{100} \right) \left(1 + \frac{r_2}{100} \right)$

$$\text{Amount} = 25000 \left(1 + \frac{4}{100} \right) \left(1 + \frac{5}{100} \right)$$

$$= 25000 \times \frac{104}{100} \times \frac{105}{100}$$

$$= 25000 \times \frac{26}{25} \times \frac{21}{20} = \text{Rs. } 27300$$

39. (c); Amount = $P \left(1 + \frac{r}{100} \right)^2$

$$1348.32 = 1200 \left(1 + \frac{r}{100} \right)^2$$

$$\frac{1348.32}{1200} = \left(1 + \frac{r}{100} \right)^2$$

$$\frac{134832}{120000} = \left(1 + \frac{r}{100} \right)^2$$

$$\frac{44944}{40000} = \left(1 + \frac{r}{100} \right)^2$$

$$\left(\frac{212}{200} \right)^2 = \left(1 + \frac{r}{100} \right)^2$$

$$\frac{212}{200} = \left(1 + \frac{r}{100} \right)$$

$$\frac{12}{200} = \frac{r}{100} \Rightarrow r = 6\%$$

40. (a); Amount = $P \left(1 + \frac{r}{100} \right)^t$

$$t = \text{number of quarters} = \frac{9}{3} = 3$$

$$\text{Amount} = 12000 \left(1 + \frac{\frac{20}{4}}{100} \right)^3$$

$$\text{Amount} = 12000 \left(\frac{105}{100} \right)^3$$

$$\text{Amount} = \text{Rs. } 13891.50$$

$$\text{CI} = 13891.50 - 12000 = \text{Rs. } 1891.50$$